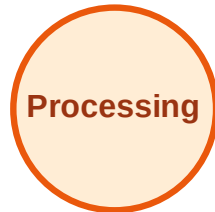


# BFS Traversal

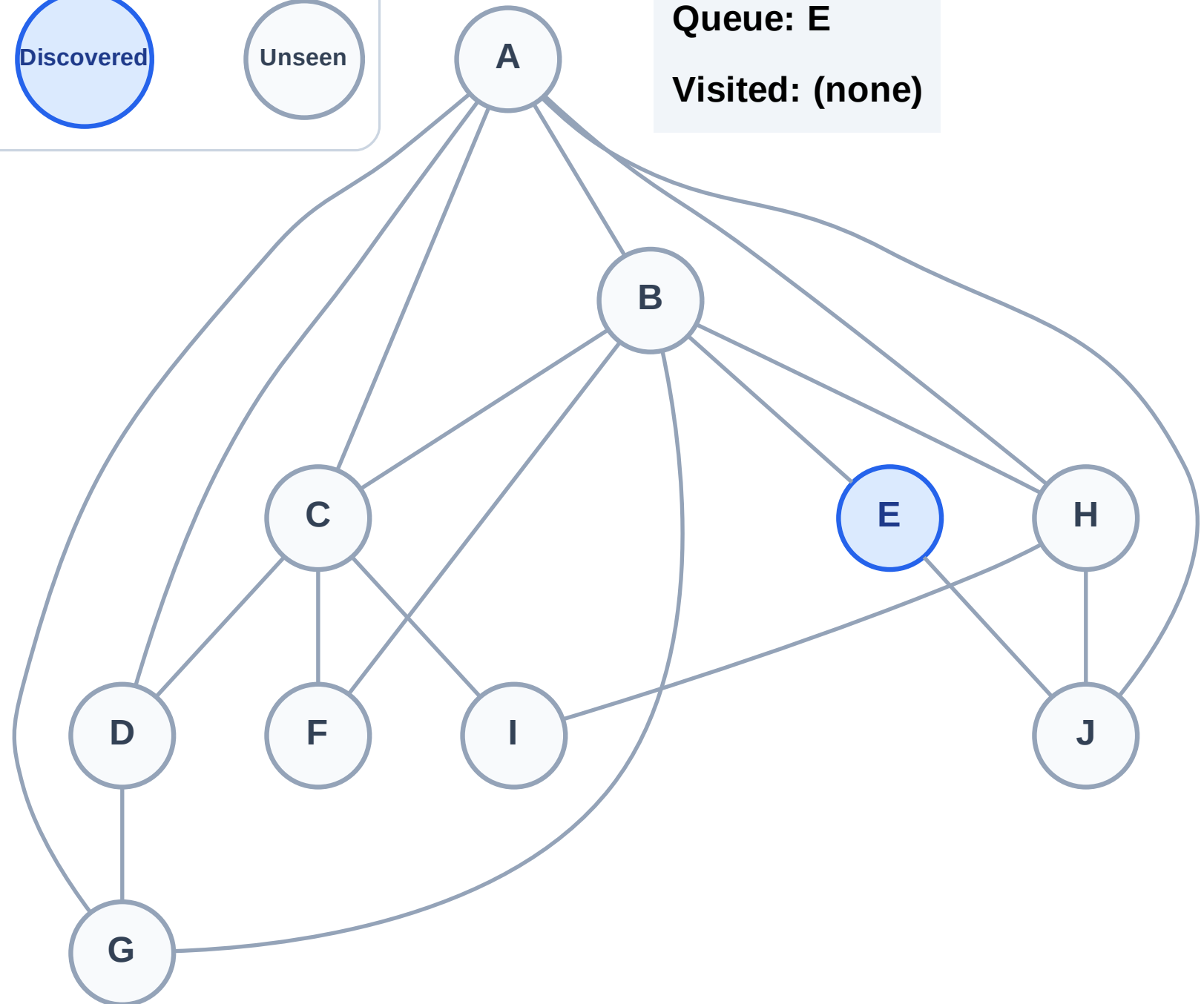
Step 1: Start at E

## Legend



Queue: E

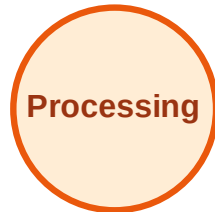
Visited: (none)



# BFS Traversal

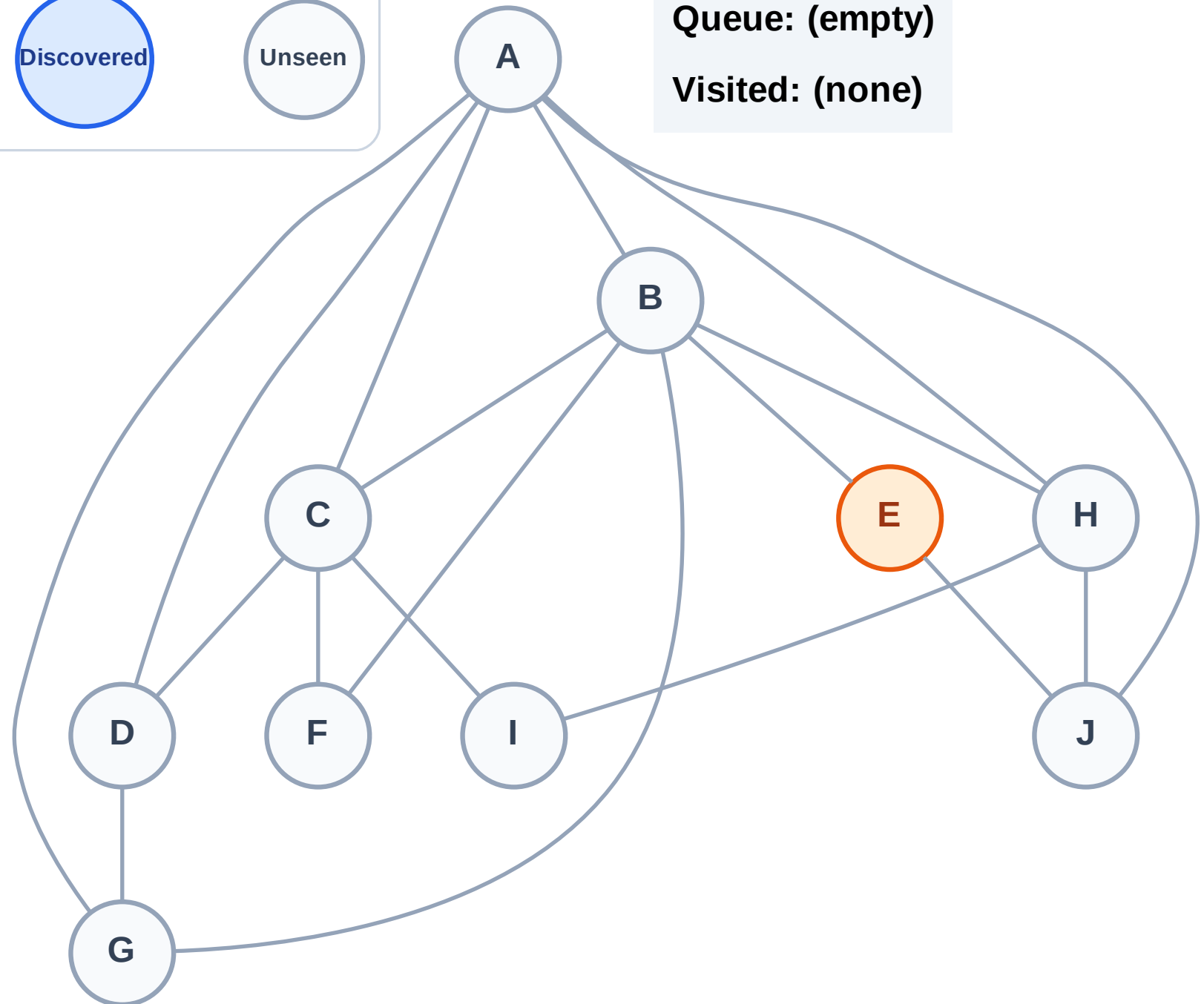
Step 2: Process node E

## Legend



Queue: (empty)

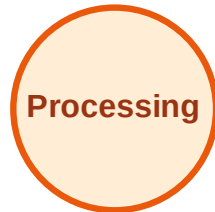
Visited: (none)



# BFS Traversal

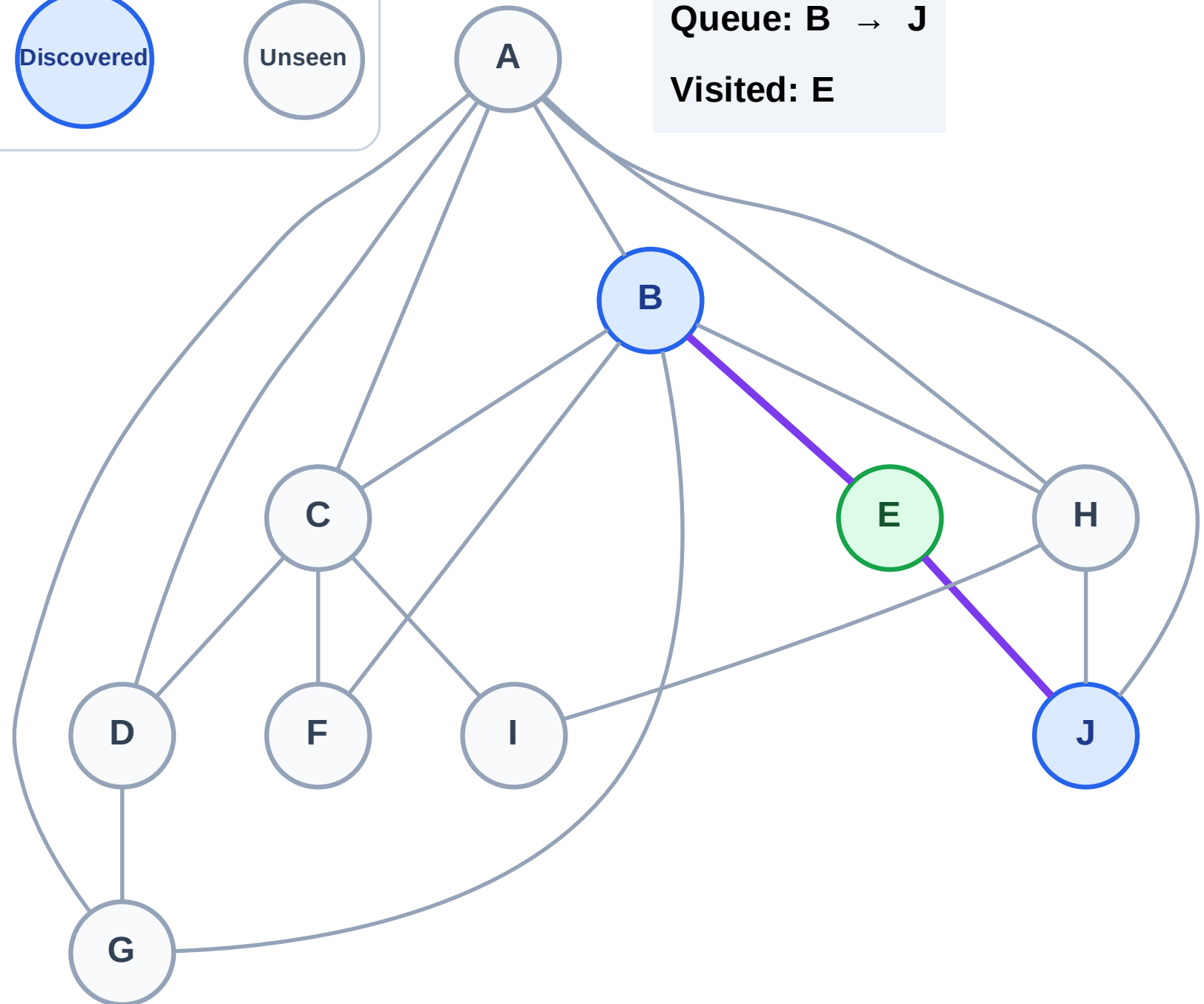
Step 3: Discover B, J from E

## Legend



Queue: B → J

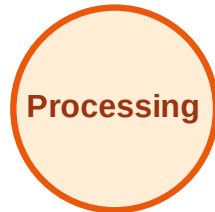
Visited: E



# BFS Traversal

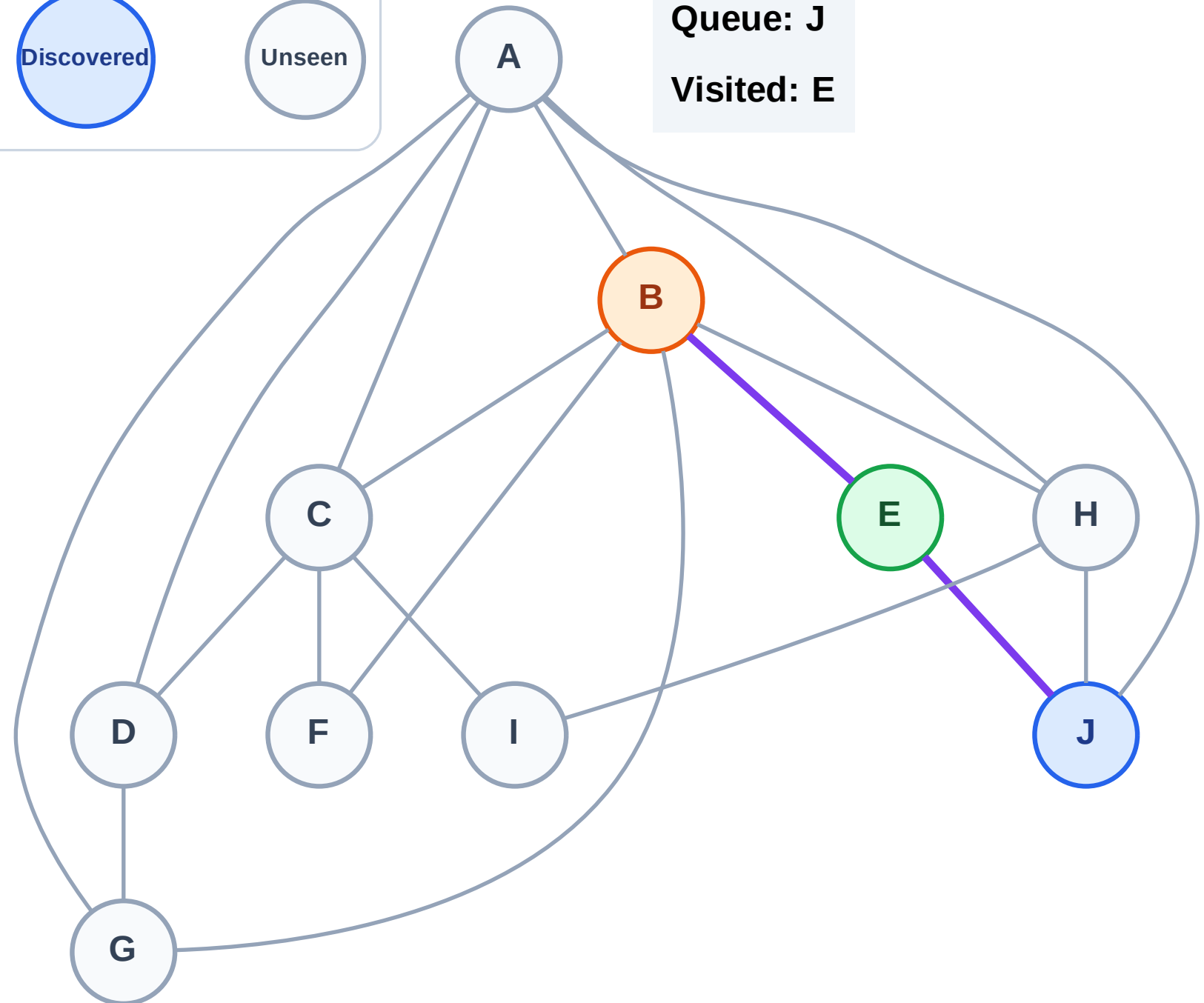
Step 4: Process node B

## Legend



Queue: J

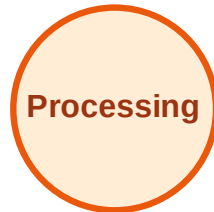
Visited: E



# BFS Traversal

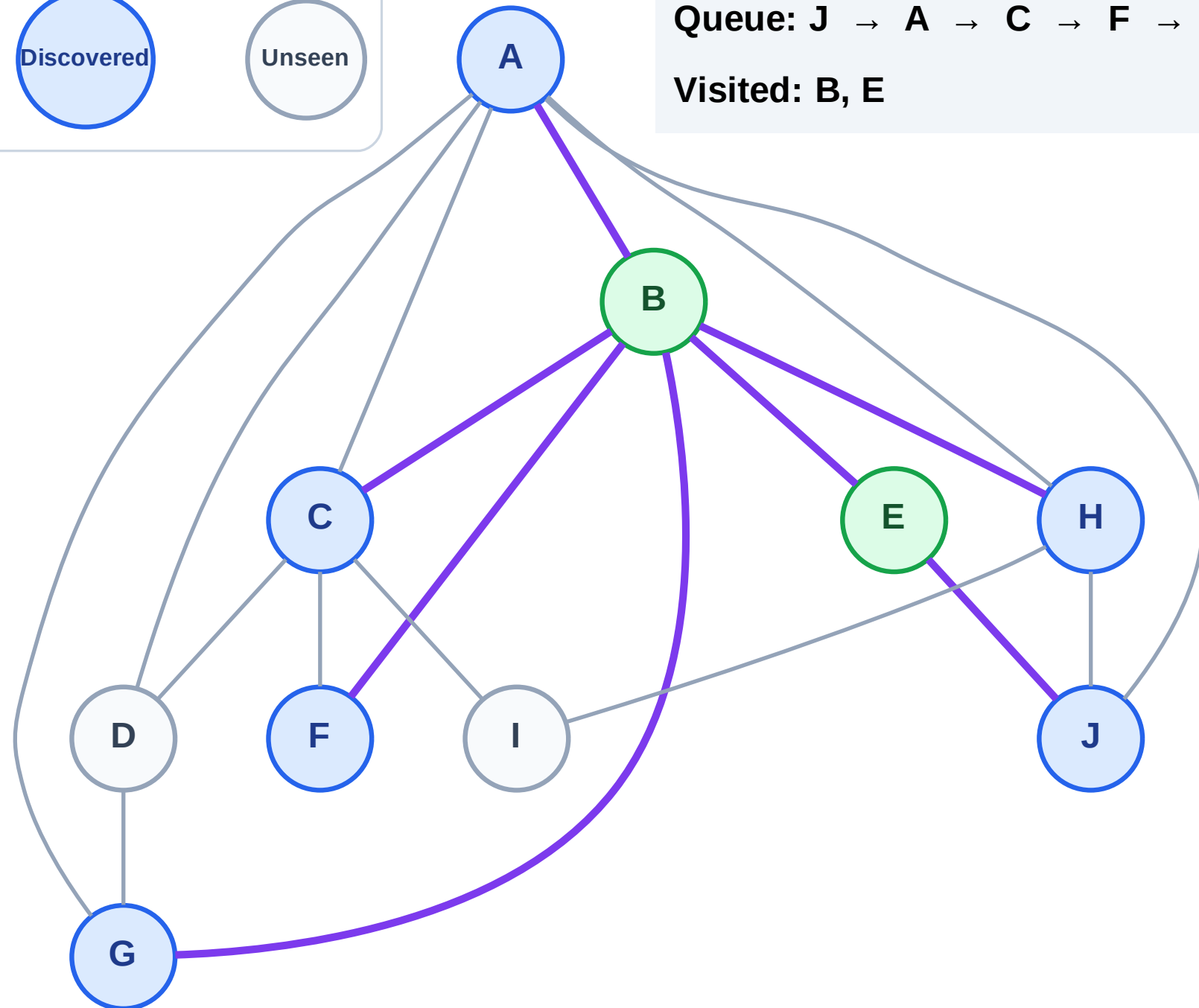
Step 5: Discover A, C, F, G, H from B

## Legend



Queue: J → A → C → F → G → H

Visited: B, E



# BFS Traversal

Step 6: Process node J

## Legend

Complete

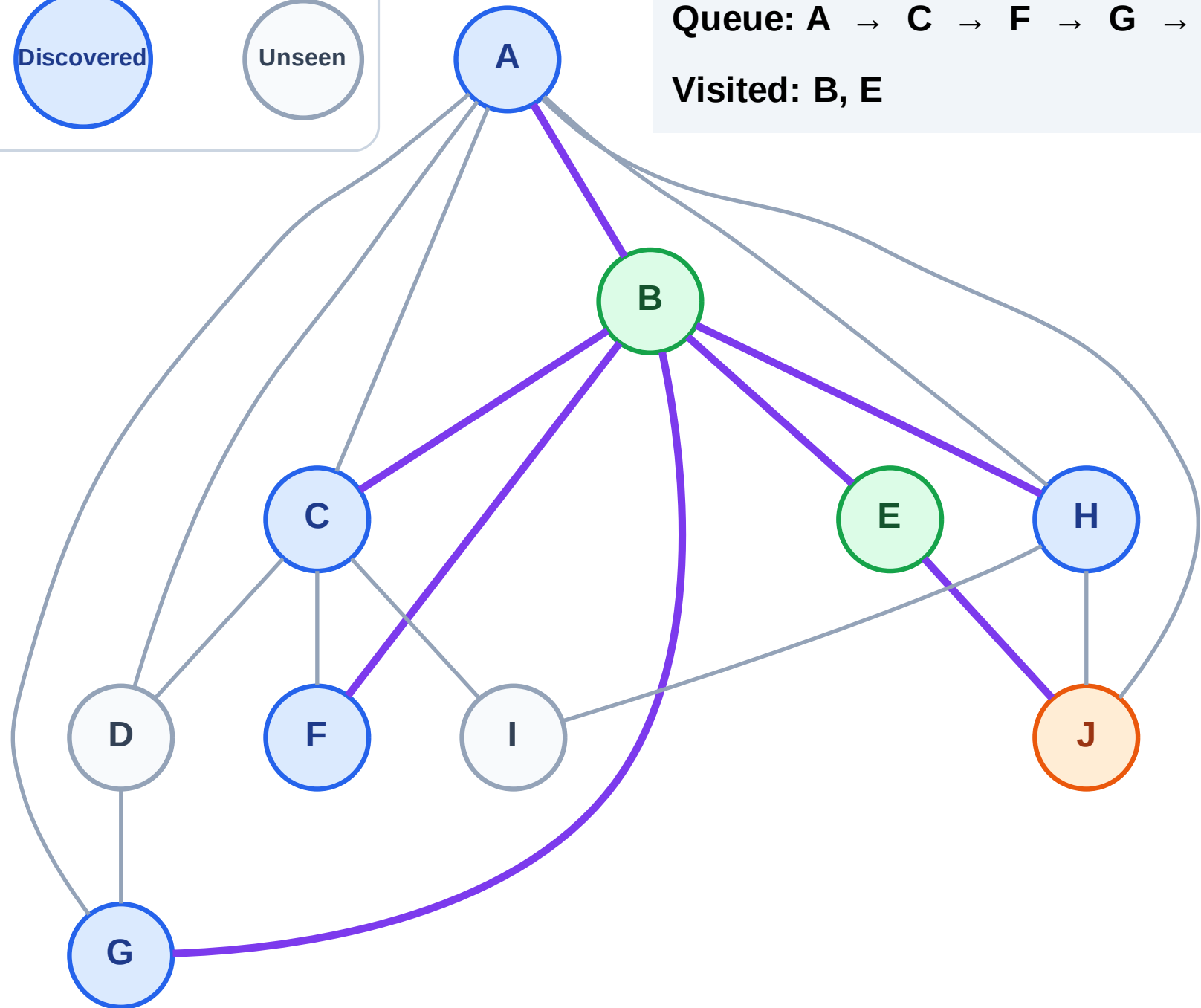
Processing

Discovered

Unseen

Queue: A → C → F → G → H

Visited: B, E



# BFS Traversal

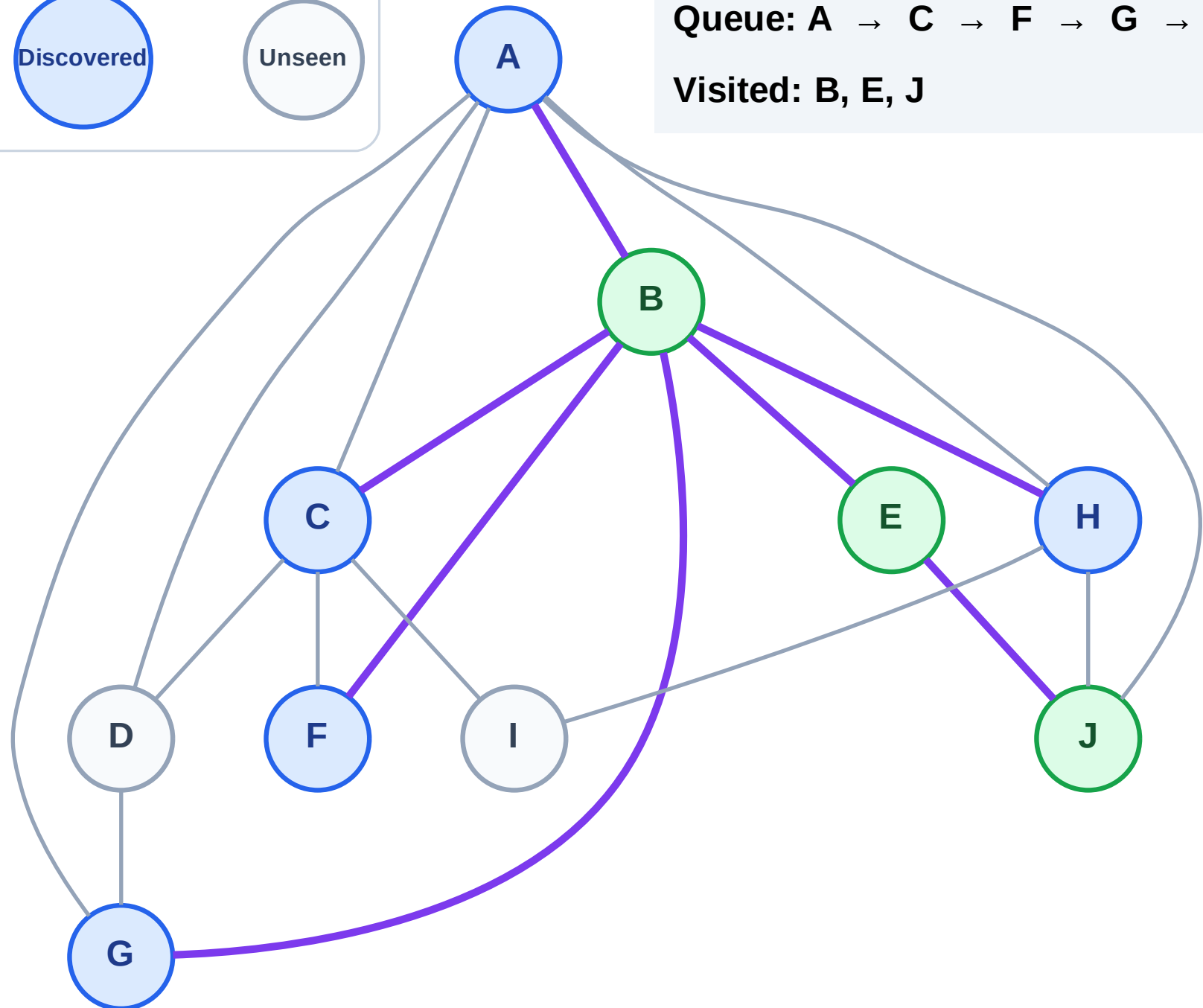
Step 7: No new neighbors from J

## Legend



Queue: A → C → F → G → H

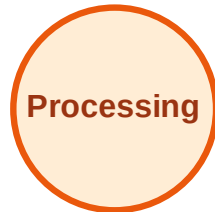
Visited: B, E, J



# BFS Traversal

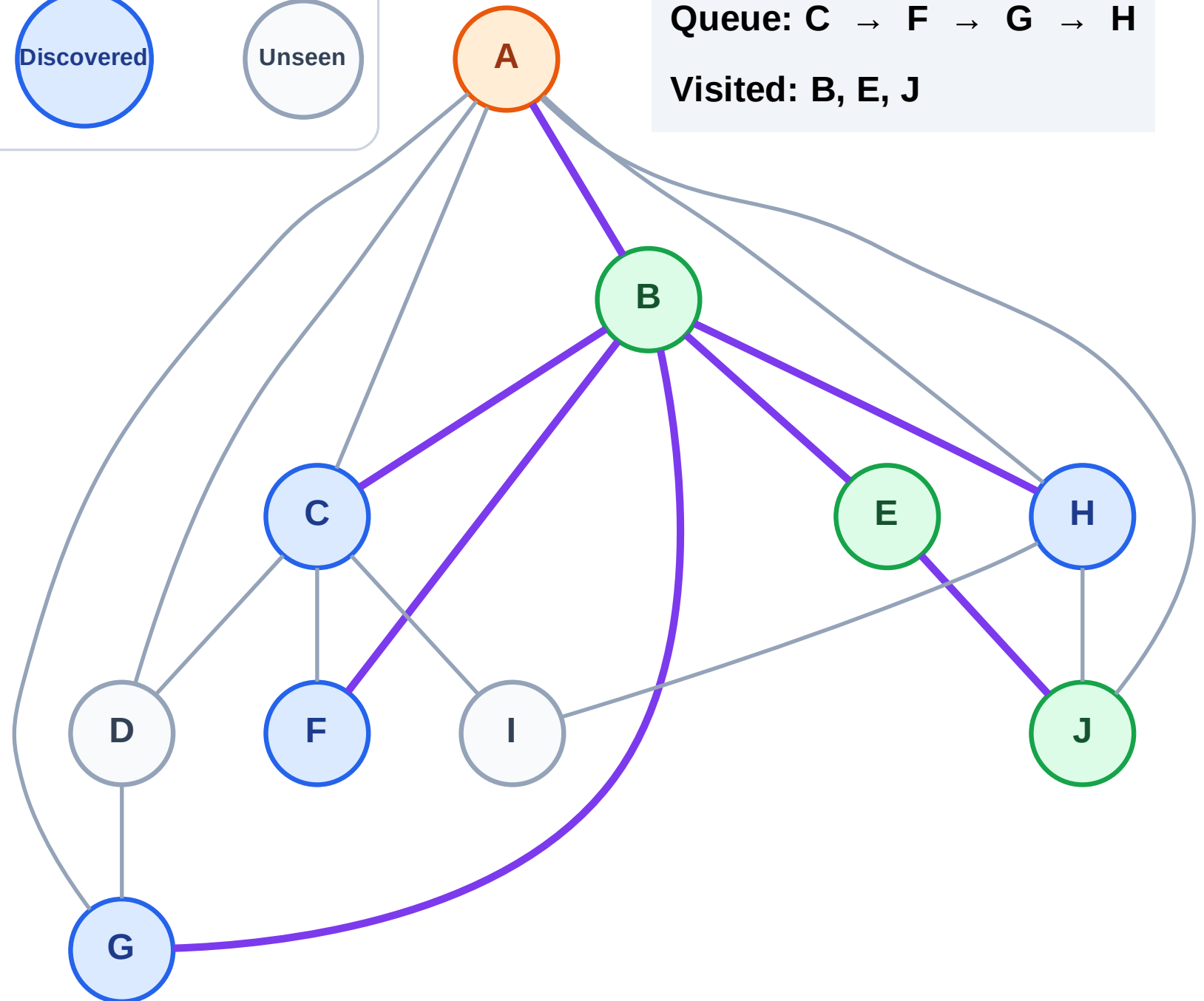
Step 8: Process node A

## Legend



Queue: C → F → G → H

Visited: B, E, J





# BFS Traversal

Step 9: Discover D from A

## Legend

Complete

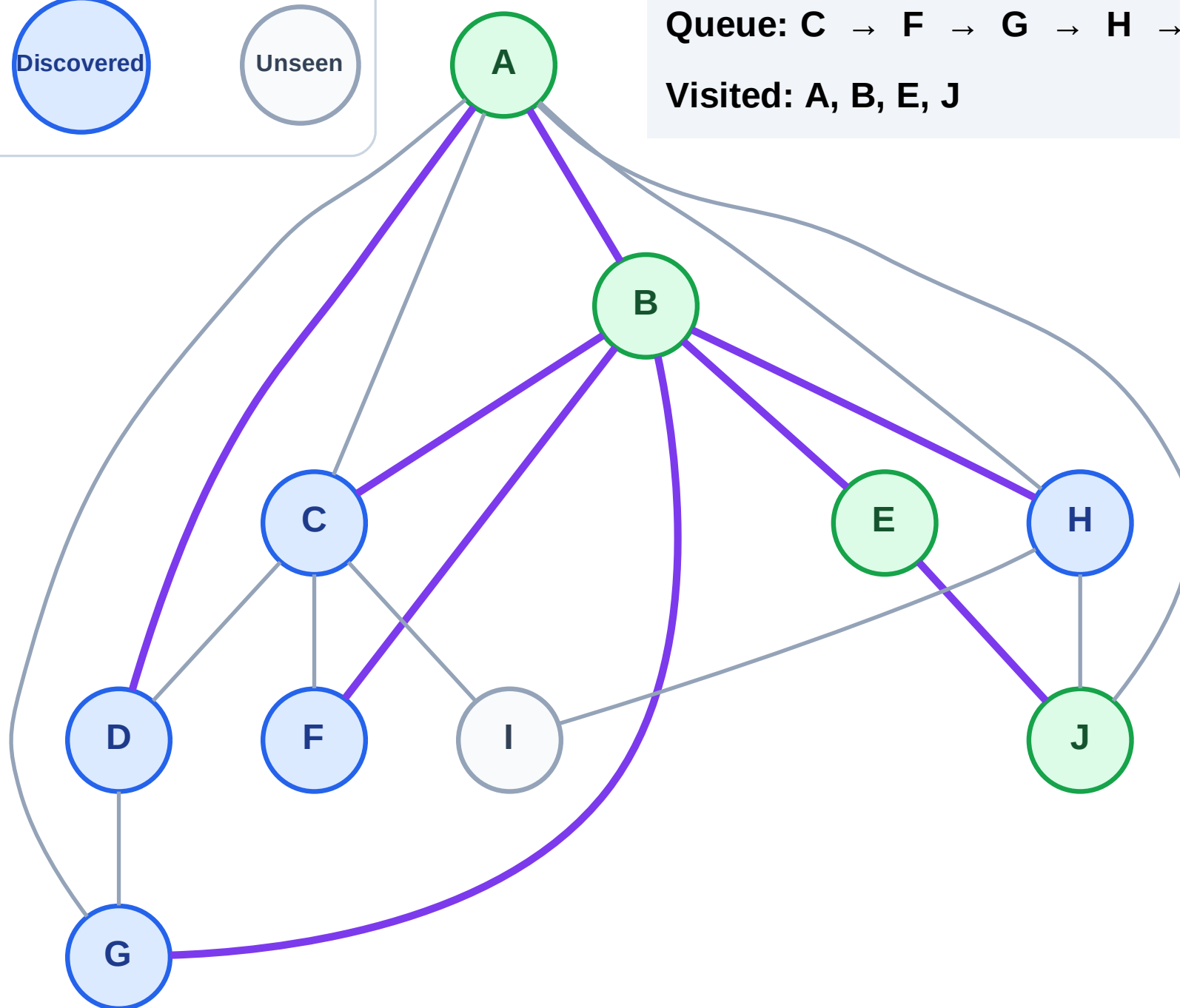
Processing

Discovered

Unseen

Queue: C → F → G → H → D

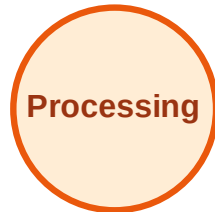
Visited: A, B, E, J



# BFS Traversal

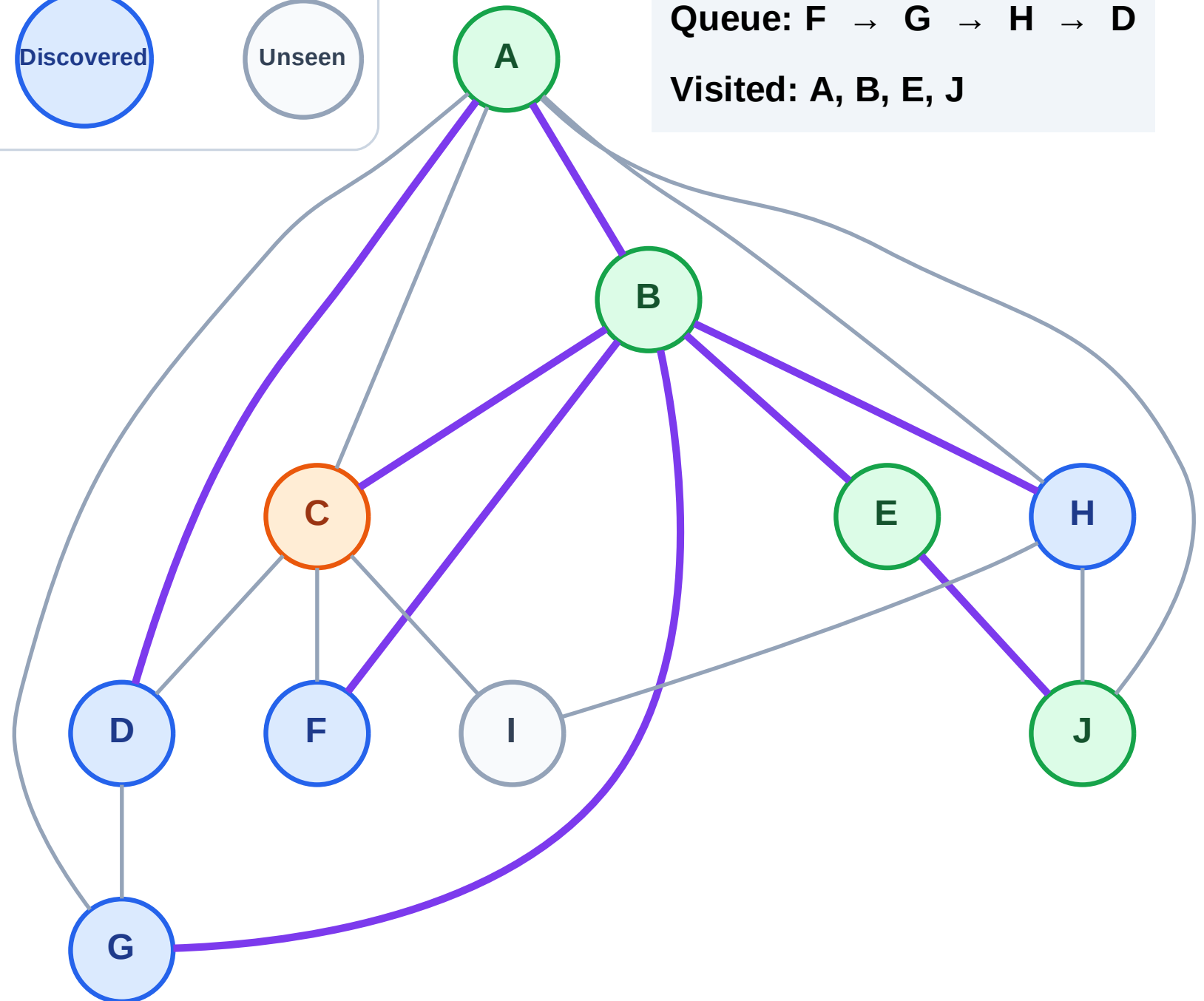
Step 10: Process node C

## Legend



Queue: F → G → H → D

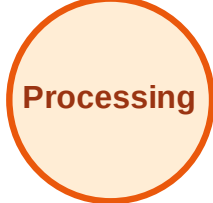
Visited: A, B, E, J



# BFS Traversal

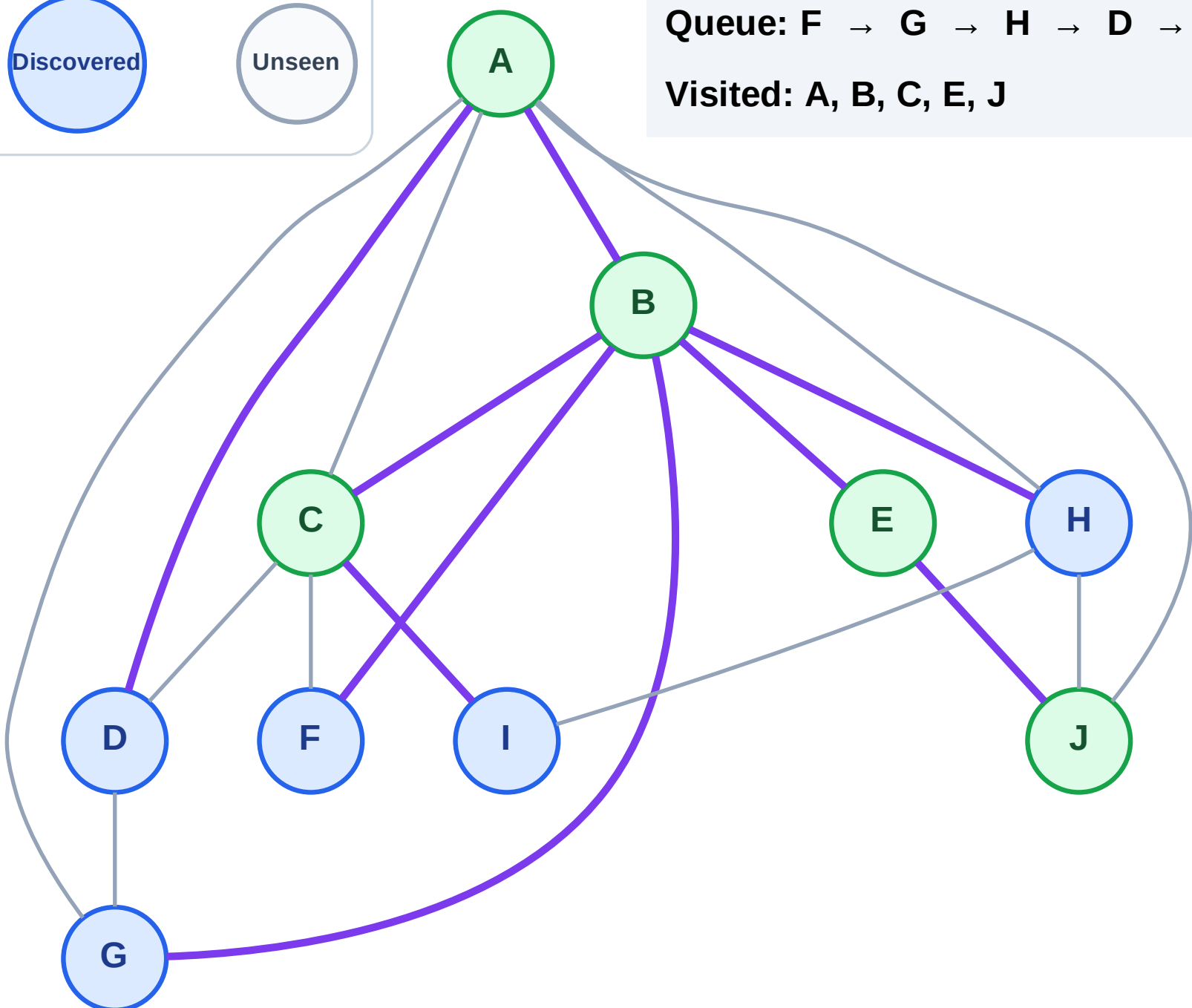
Step 11: Discover I from C

## Legend



Queue: F → G → H → D → I

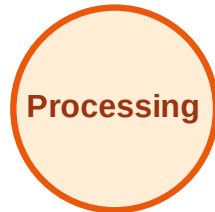
Visited: A, B, C, E, J



# BFS Traversal

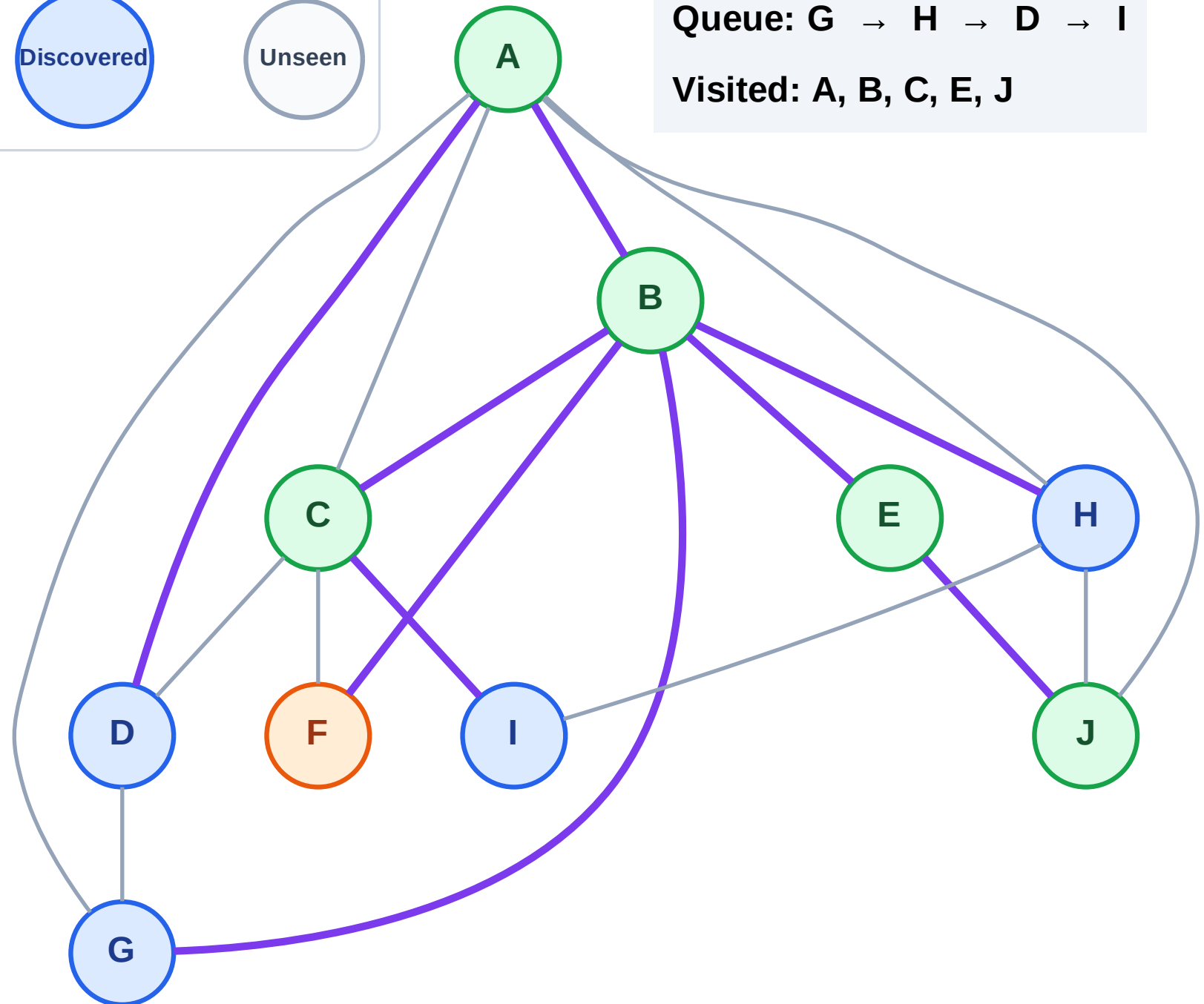
Step 12: Process node F

## Legend



Queue: G → H → D → I

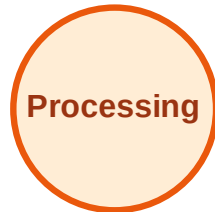
Visited: A, B, C, E, J



# BFS Traversal

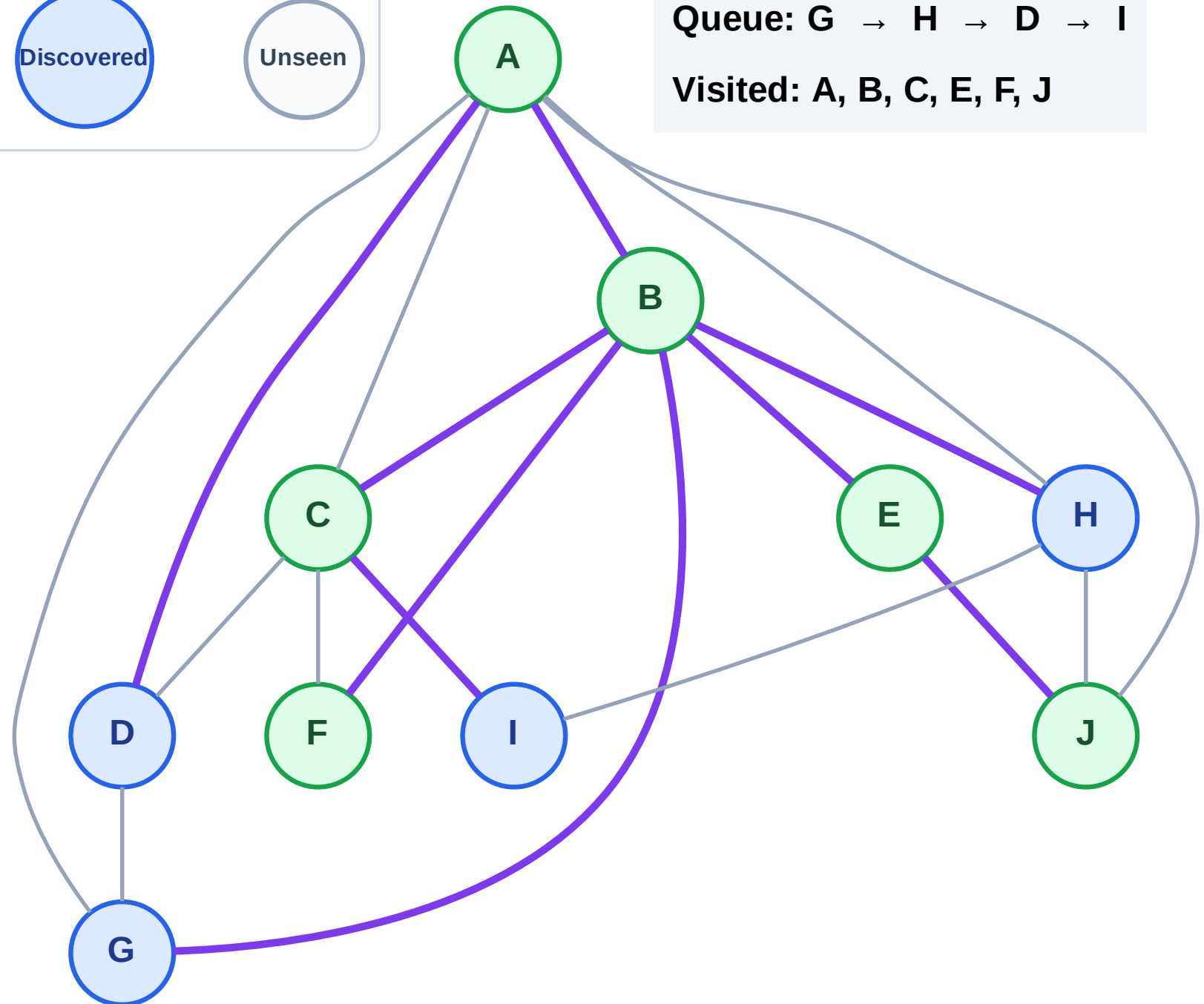
Step 13: No new neighbors from F

## Legend



Queue: G → H → D → I

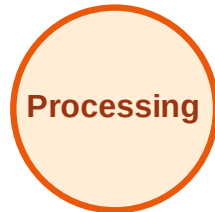
Visited: A, B, C, E, F, J



# BFS Traversal

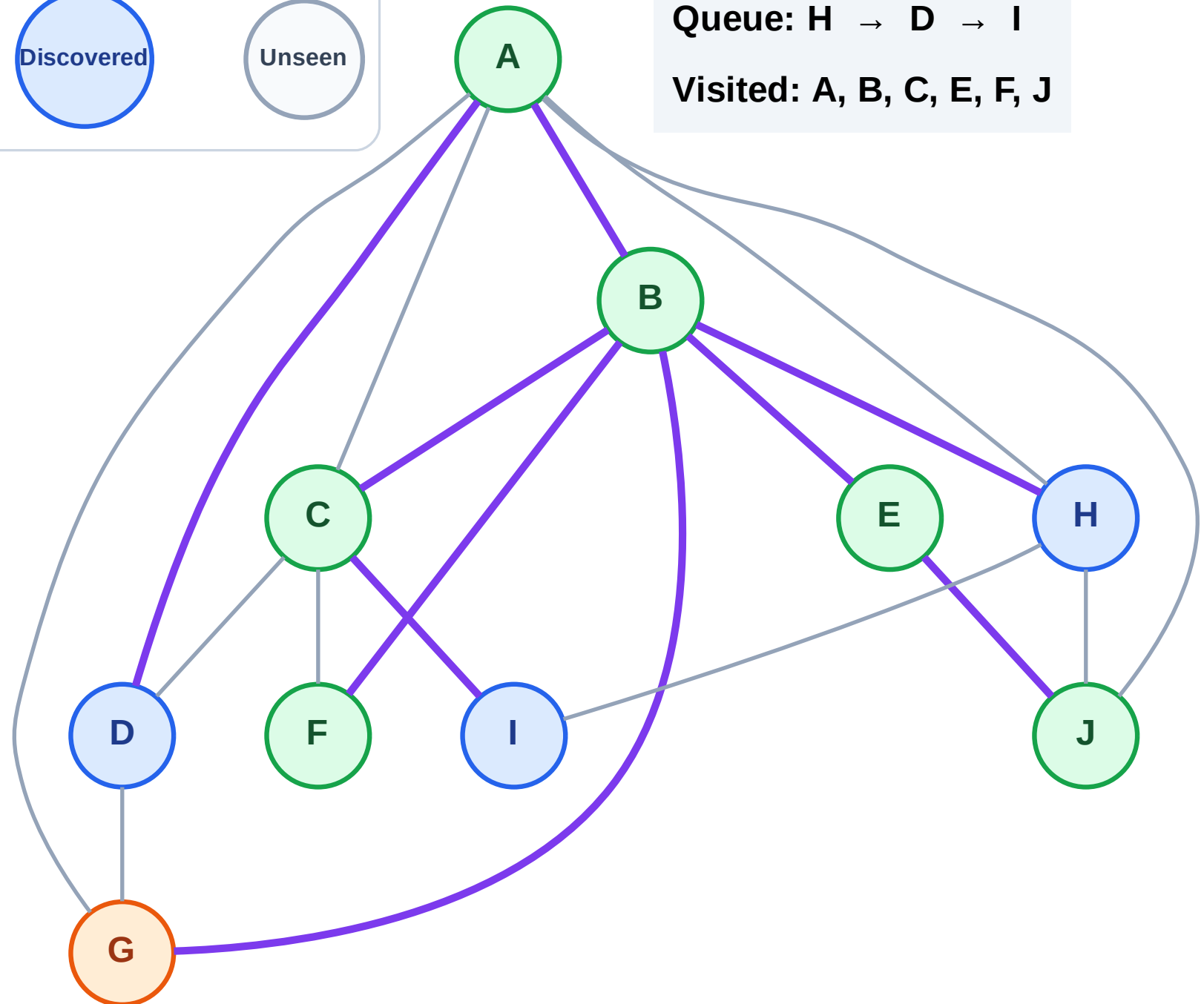
Step 14: Process node G

## Legend



Queue: H → D → I

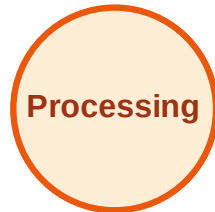
Visited: A, B, C, E, F, J



# BFS Traversal

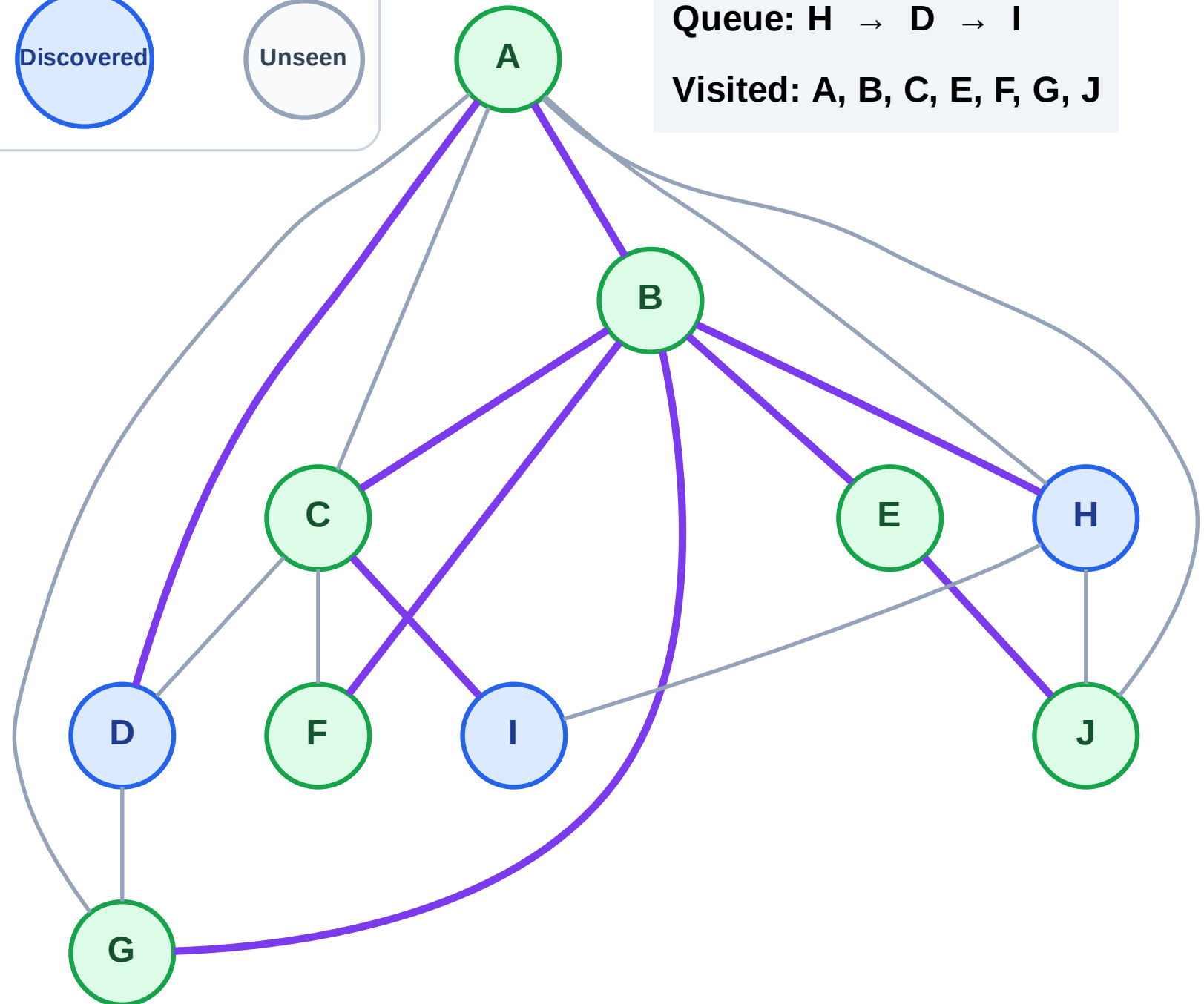
Step 15: No new neighbors from G

## Legend



Queue: H → D → I

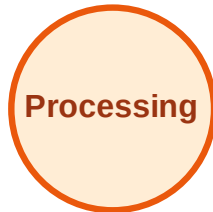
Visited: A, B, C, E, F, G, J



# BFS Traversal

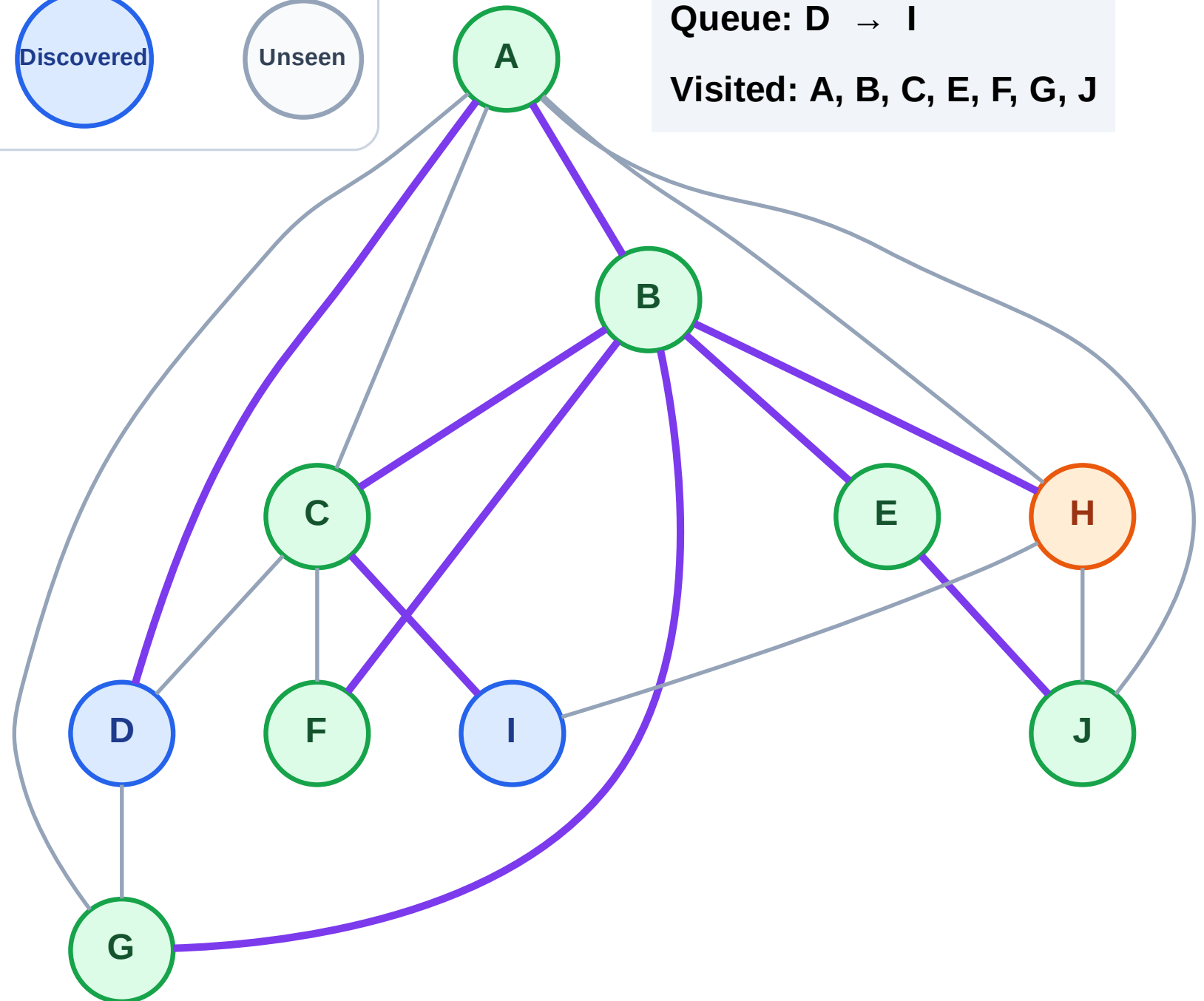
Step 16: Process node H

## Legend



Queue: D → I

Visited: A, B, C, E, F, G, J

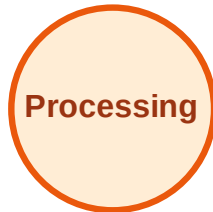




# BFS Traversal

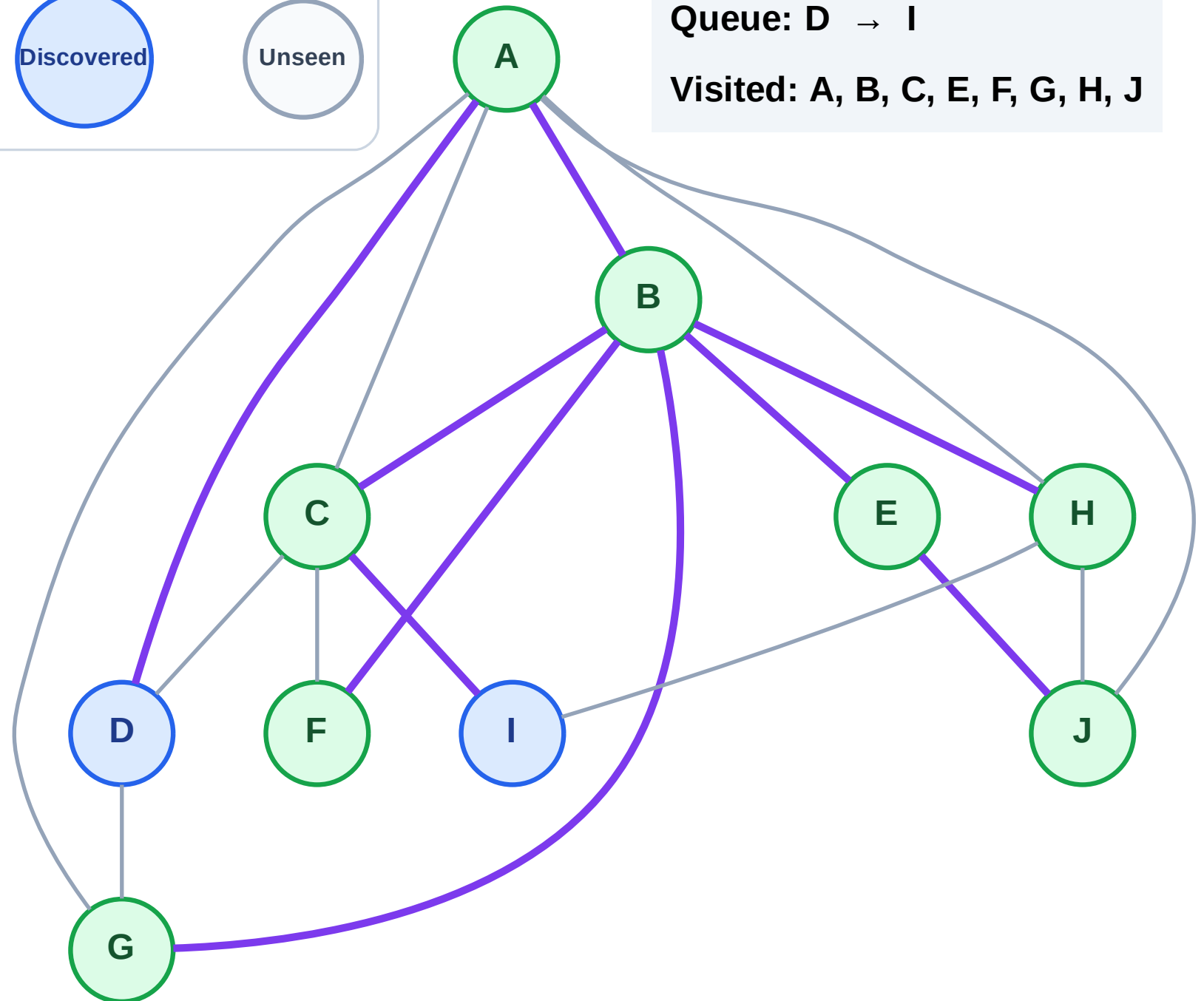
Step 17: No new neighbors from H

## Legend



Queue: D → I

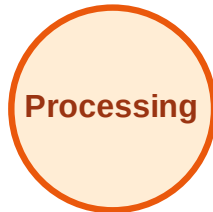
Visited: A, B, C, E, F, G, H, J



# BFS Traversal

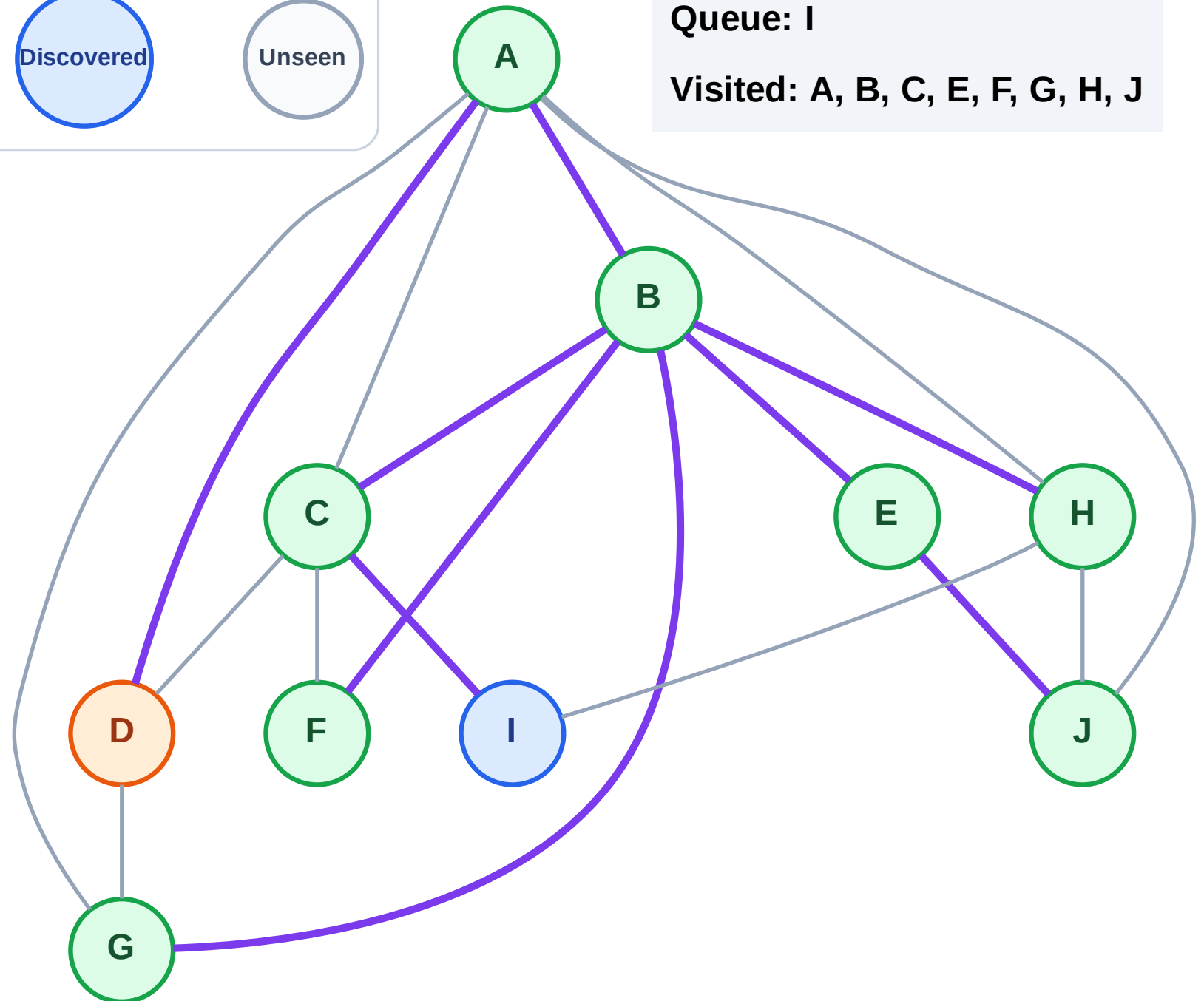
Step 18: Process node D

## Legend



Queue: I

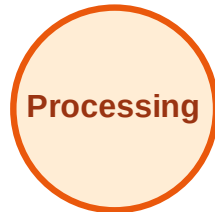
Visited: A, B, C, E, F, G, H, J



# BFS Traversal

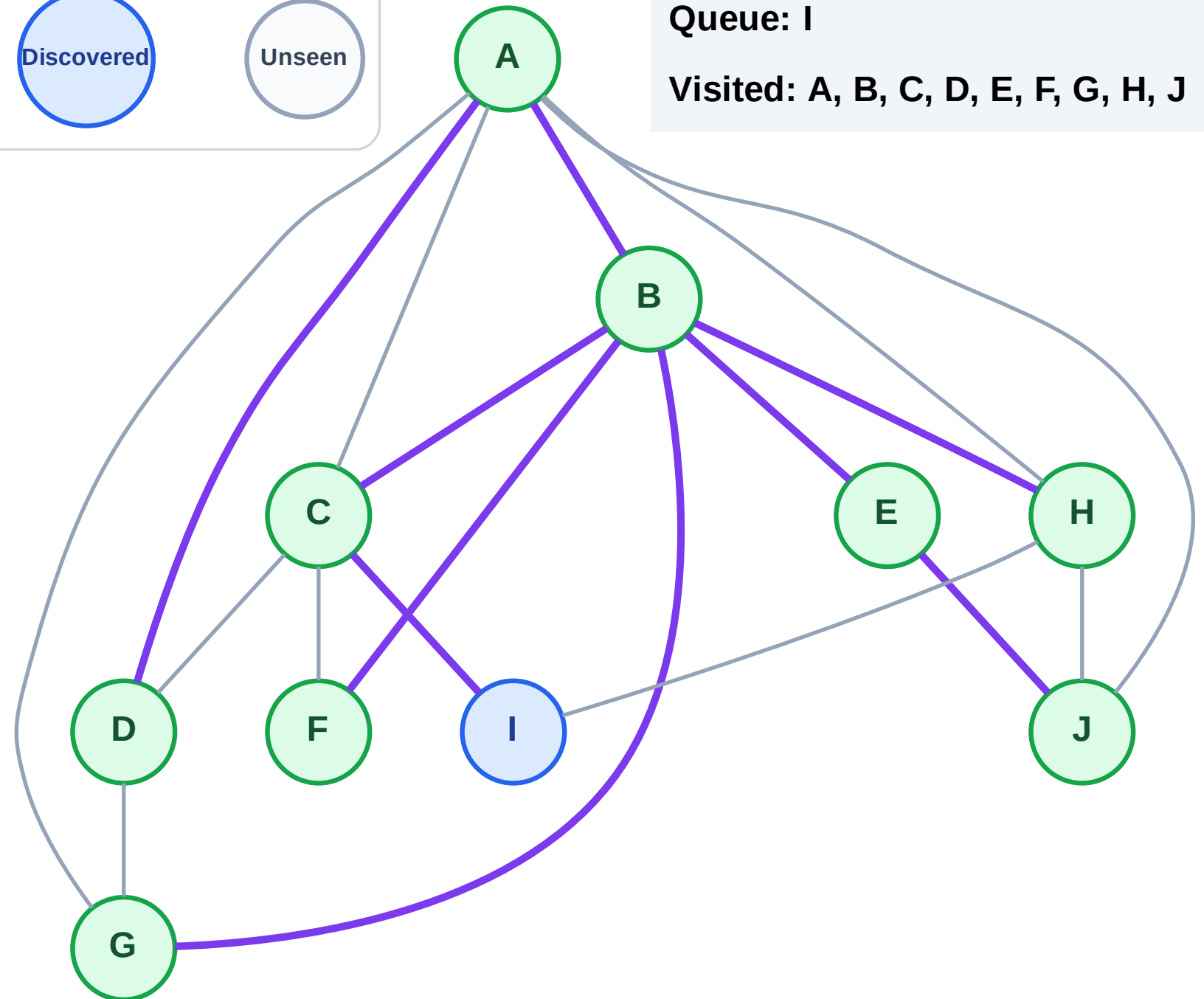
Step 19: No new neighbors from D

## Legend



Queue: I

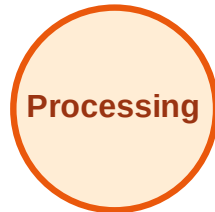
Visited: A, B, C, D, E, F, G, H, J



# BFS Traversal

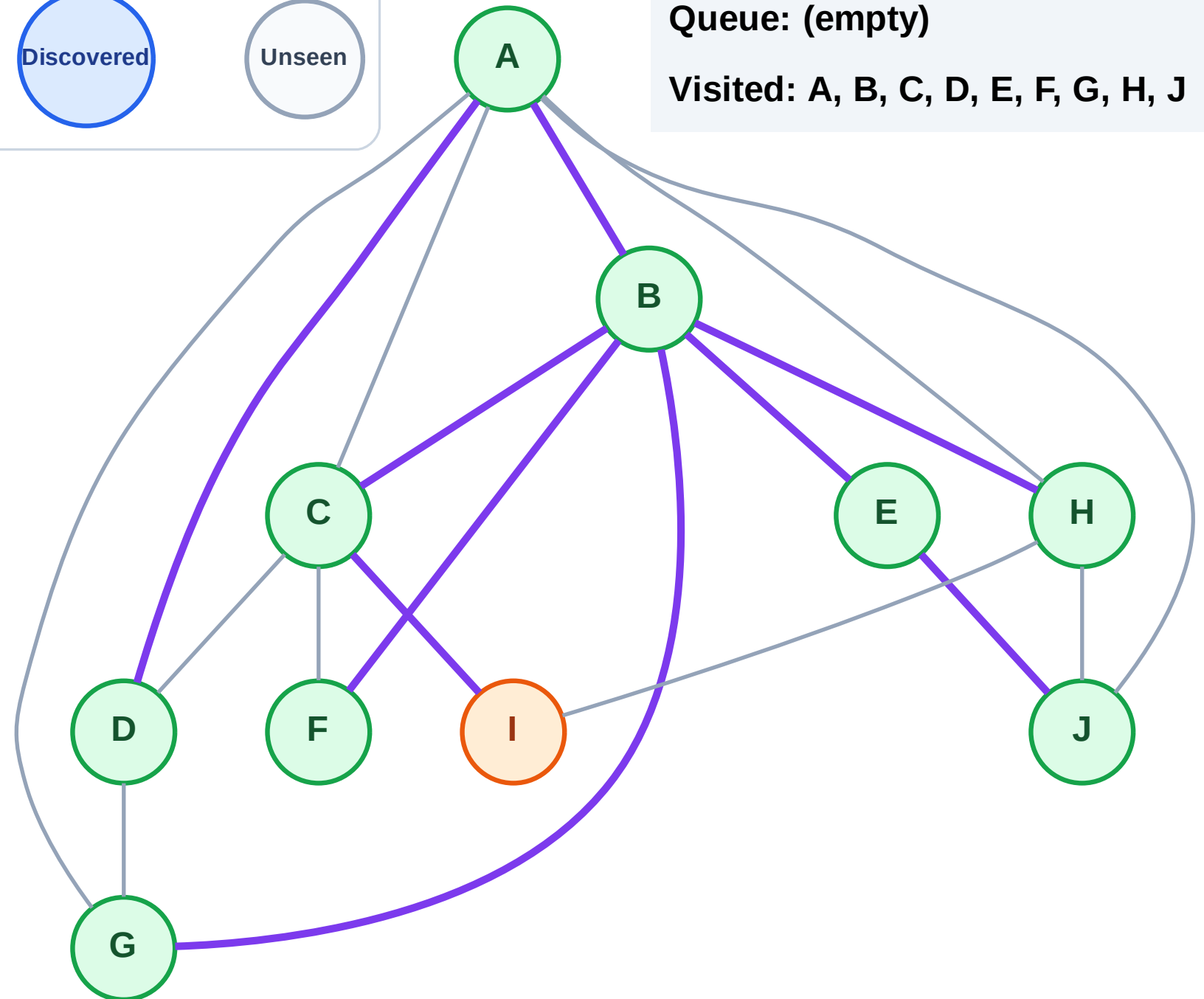
Step 20: Process node I

## Legend



Queue: (empty)

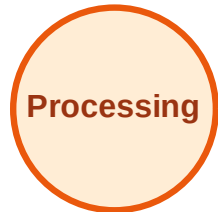
Visited: A, B, C, D, E, F, G, H, J



# BFS Traversal

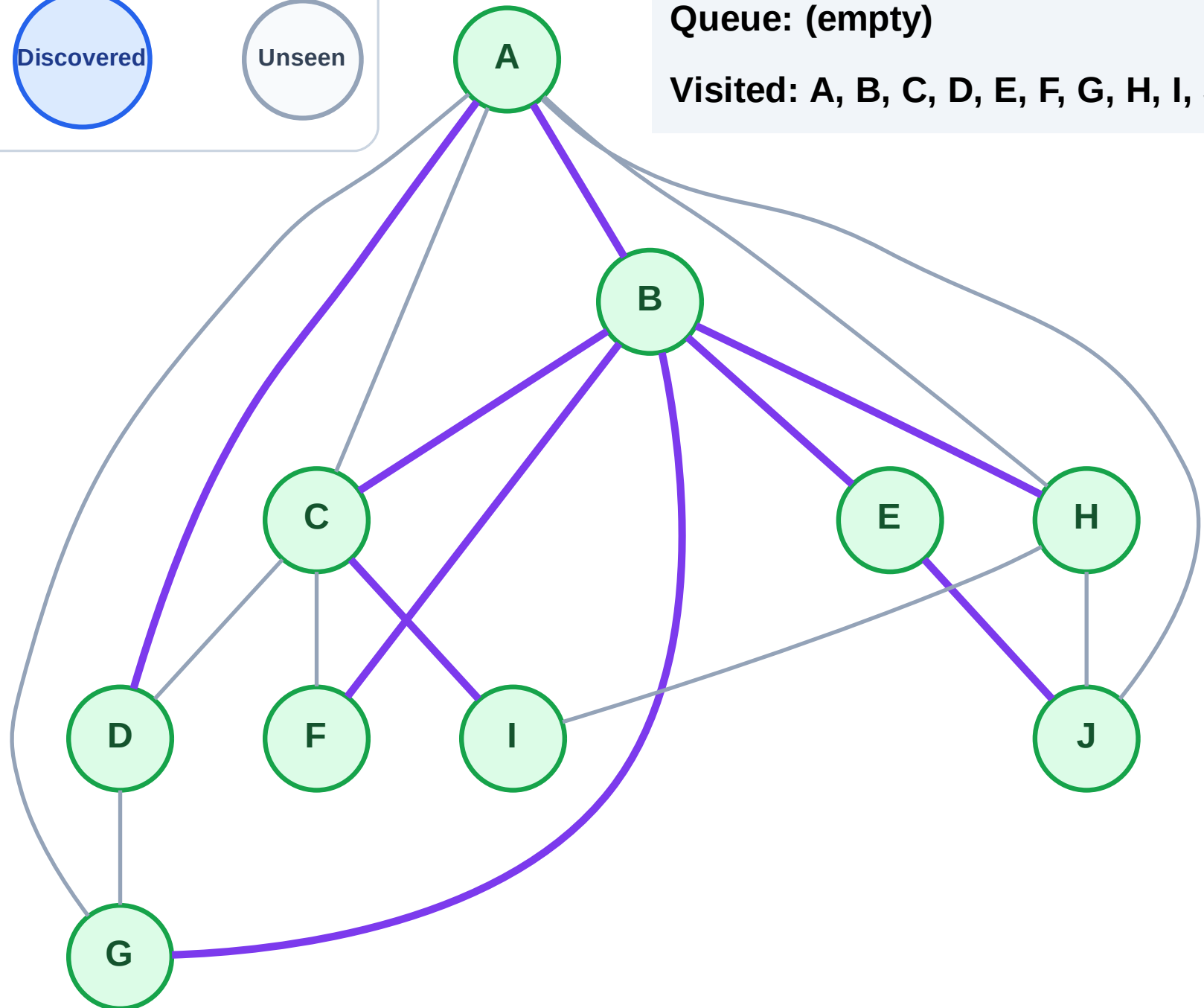
Step 21: No new neighbors from I

## Legend



Queue: (empty)

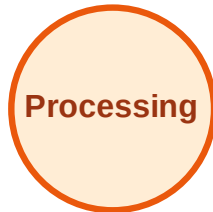
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

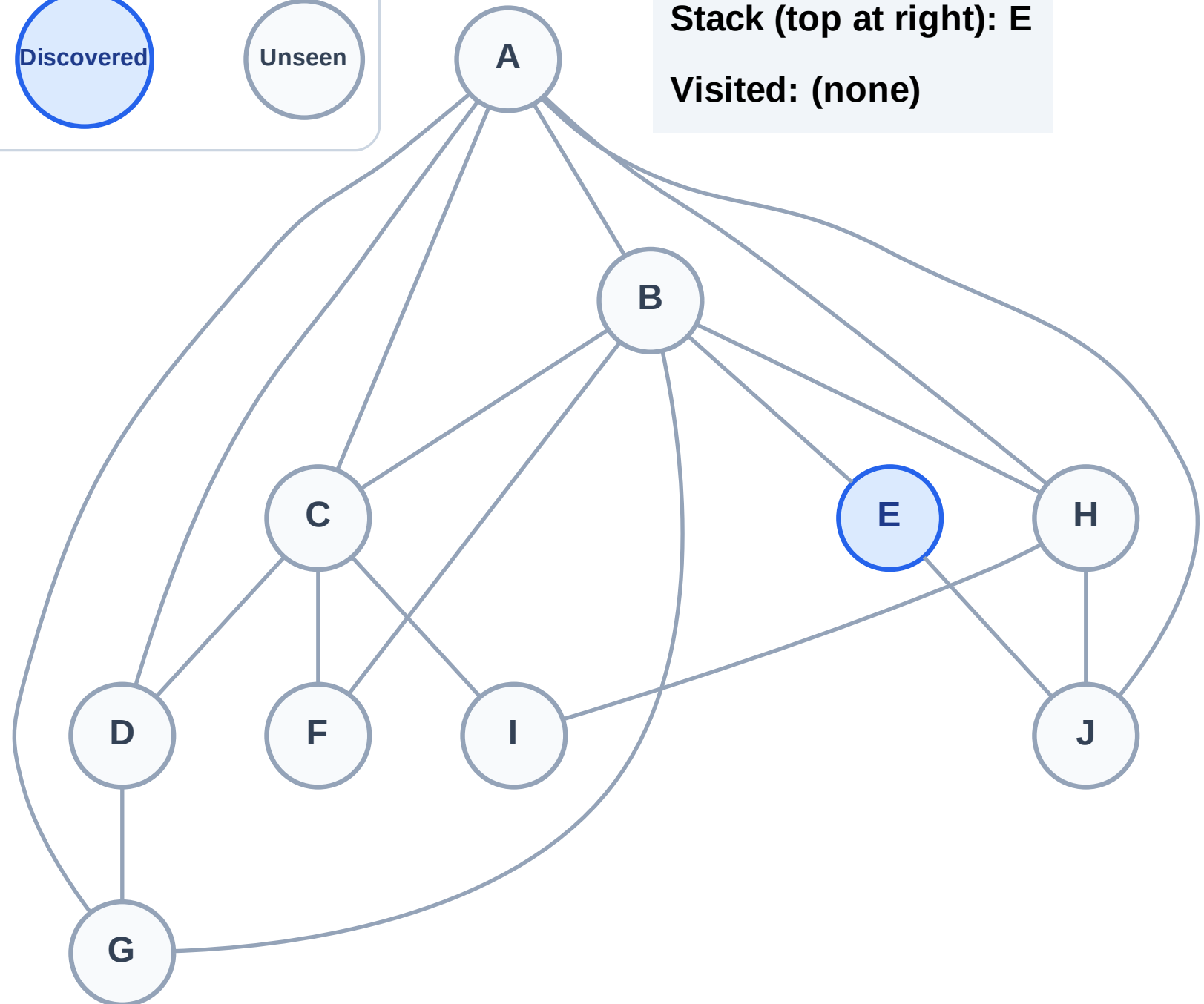
Step 1: Start at E

## Legend



Stack (top at right): E

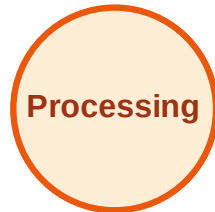
Visited: (none)



# DFS Traversal

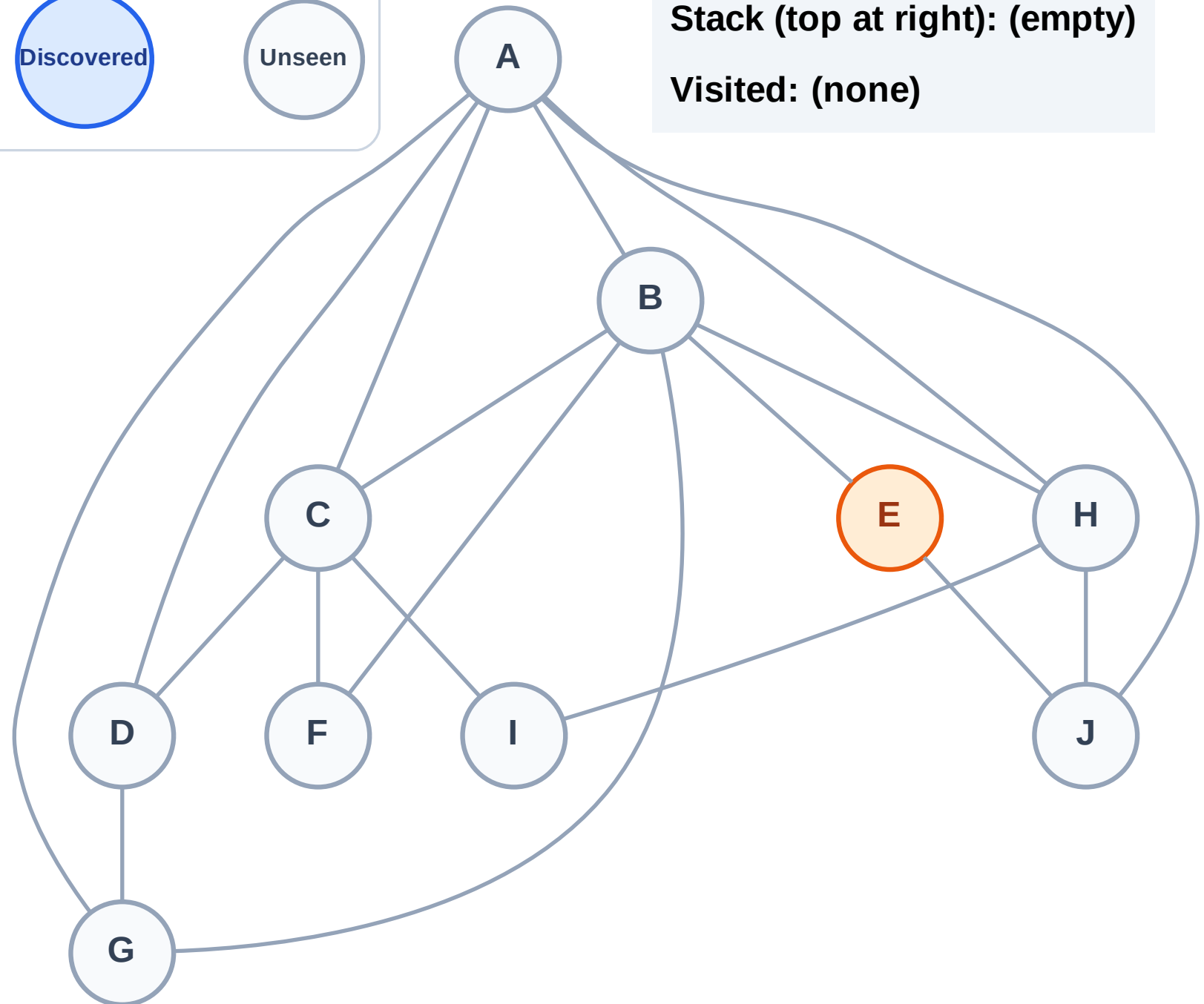
Step 2: Process node E

## Legend



Stack (top at right): (empty)

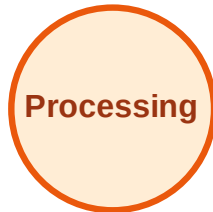
Visited: (none)



# DFS Traversal

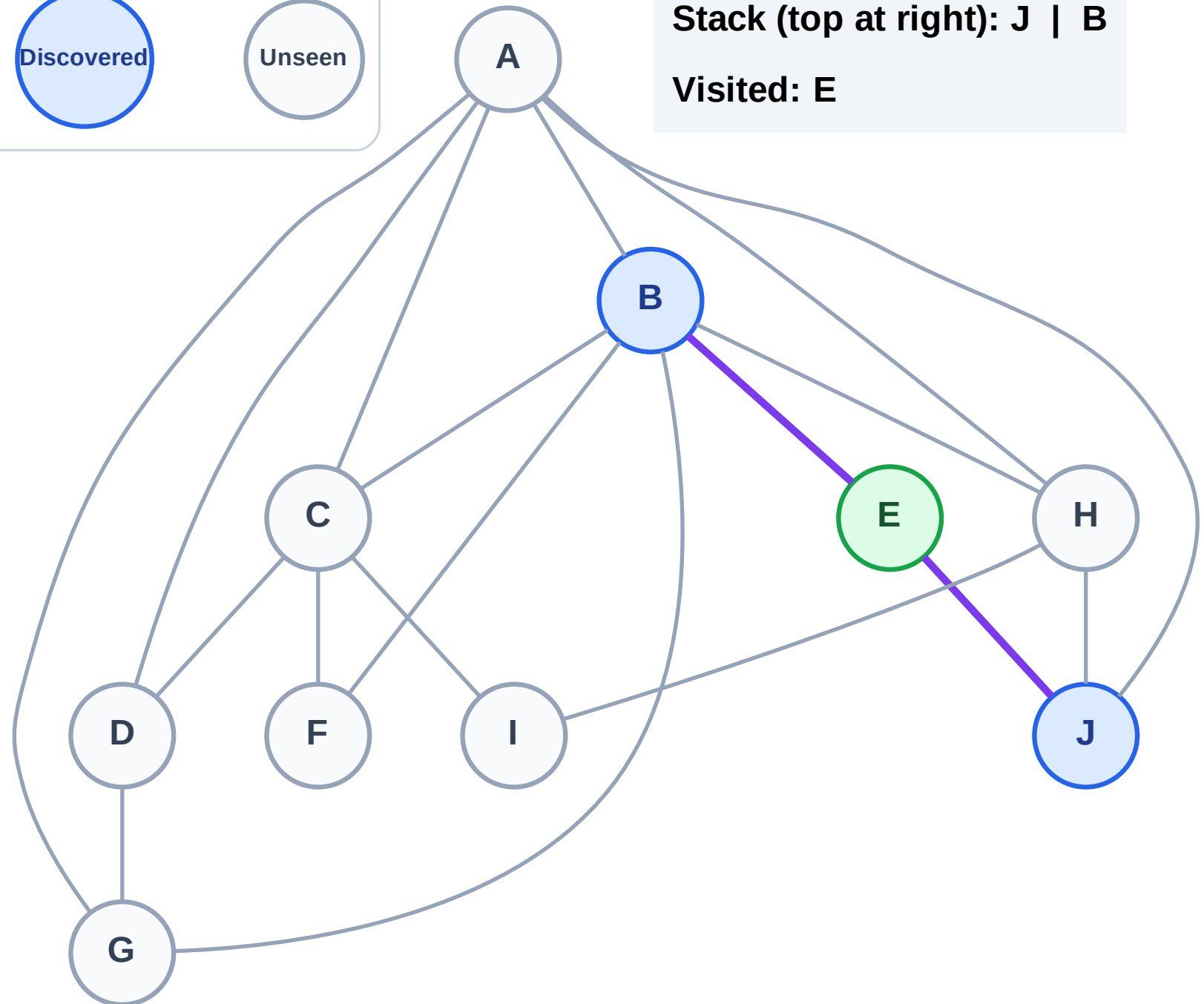
Step 3: Discover J, B from E

## Legend



Stack (top at right): J | B

Visited: E

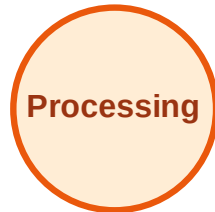




# DFS Traversal

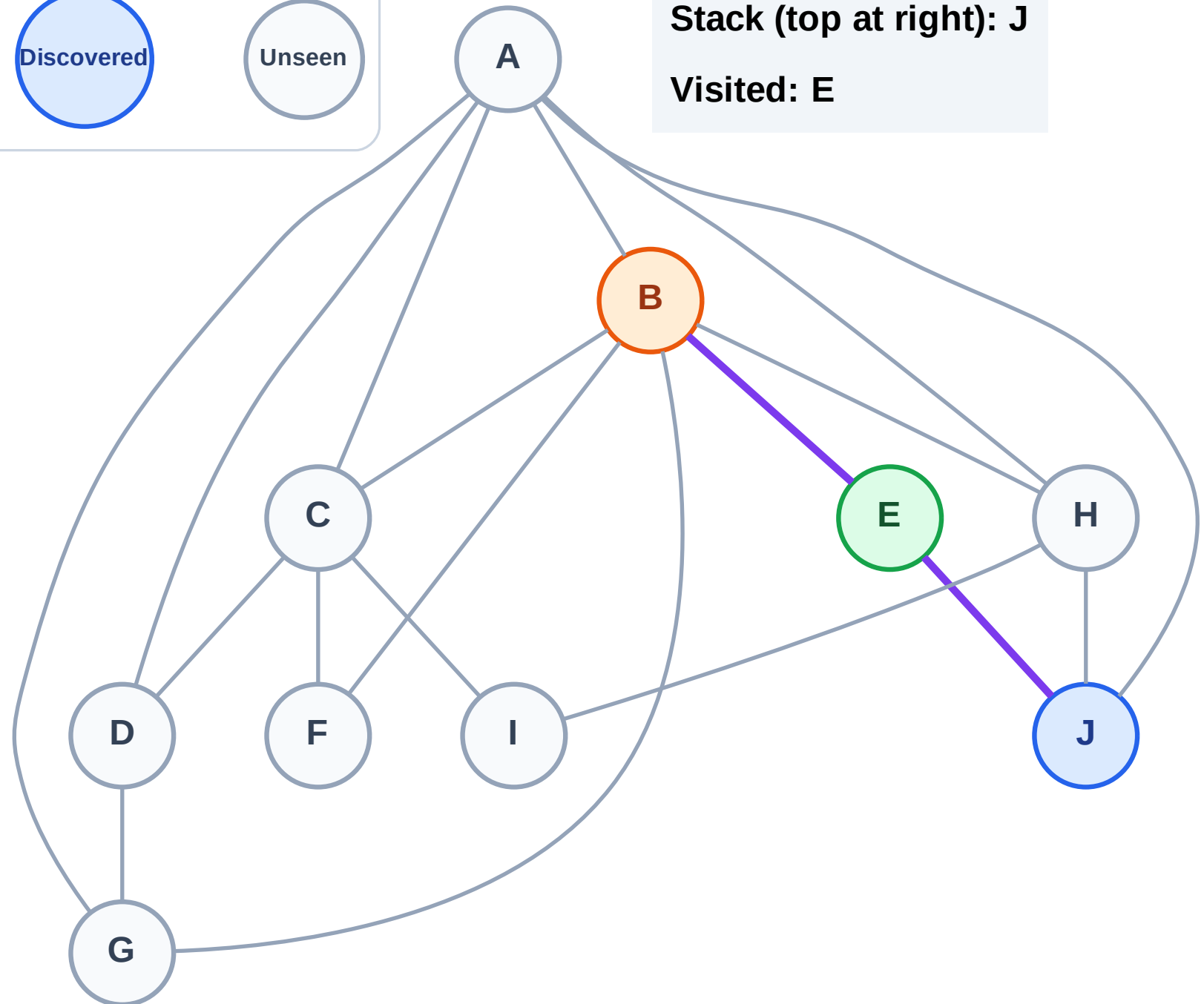
Step 4: Process node B

## Legend



Stack (top at right): J

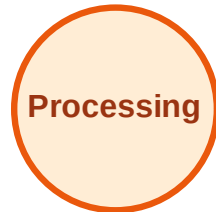
Visited: E



# DFS Traversal

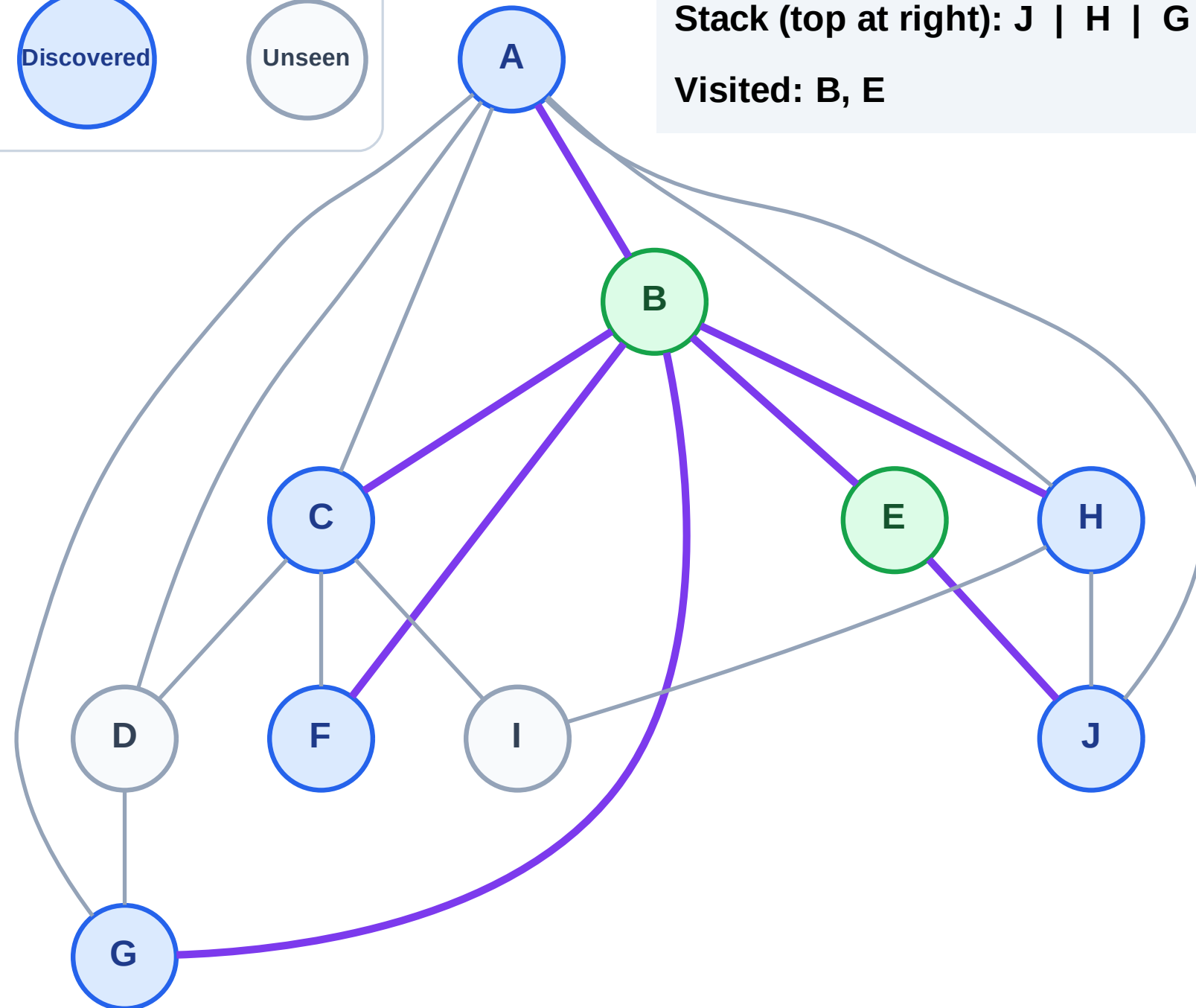
Step 5: Discover H, G, F, C, A from B

## Legend



Stack (top at right): J | H | G | F | C | A

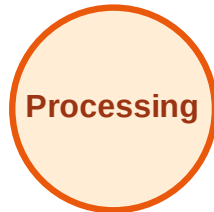
Visited: B, E



# DFS Traversal

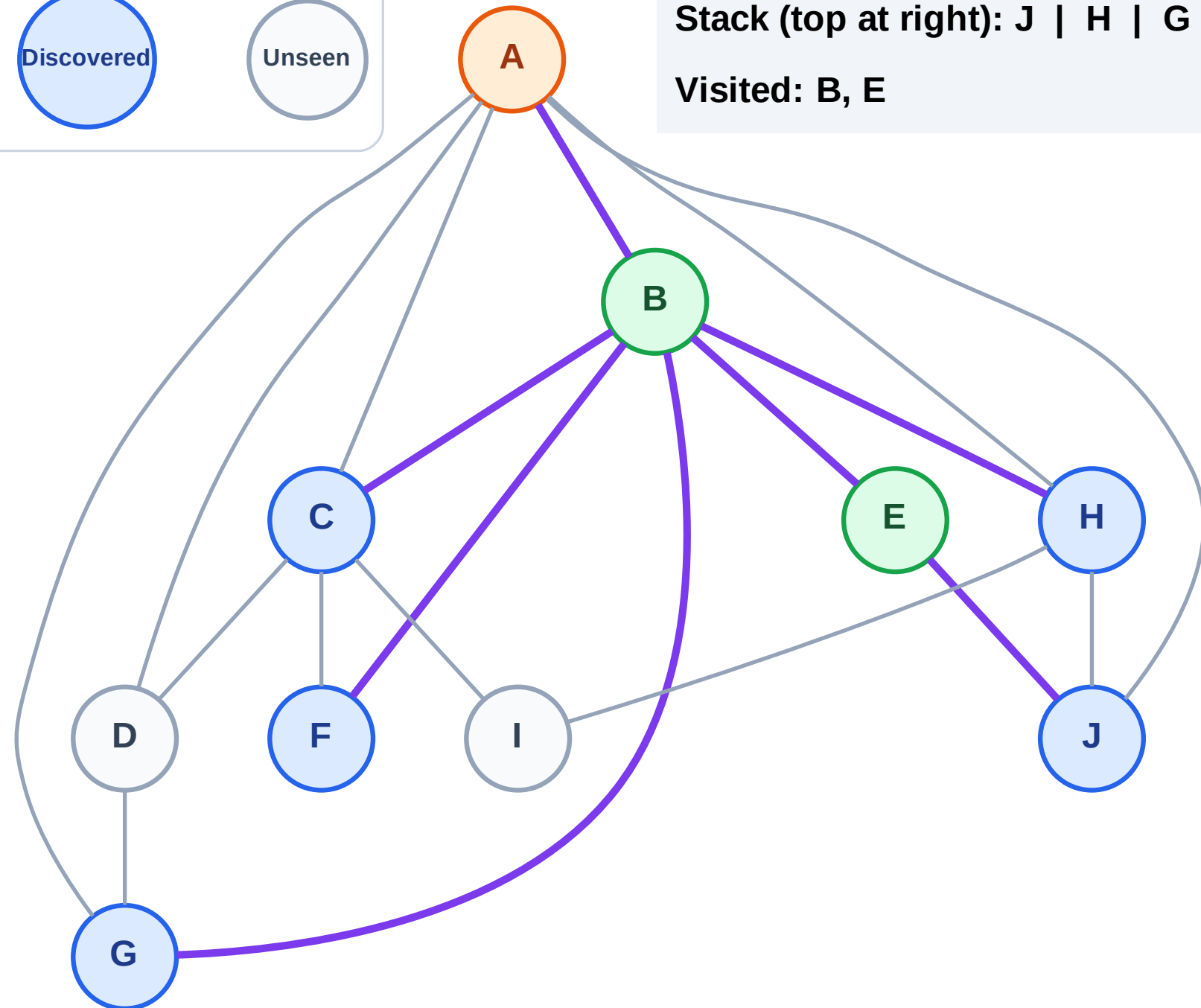
Step 6: Process node A

## Legend



Stack (top at right): J | H | G | F | C

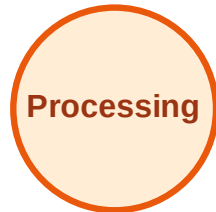
Visited: B, E



# DFS Traversal

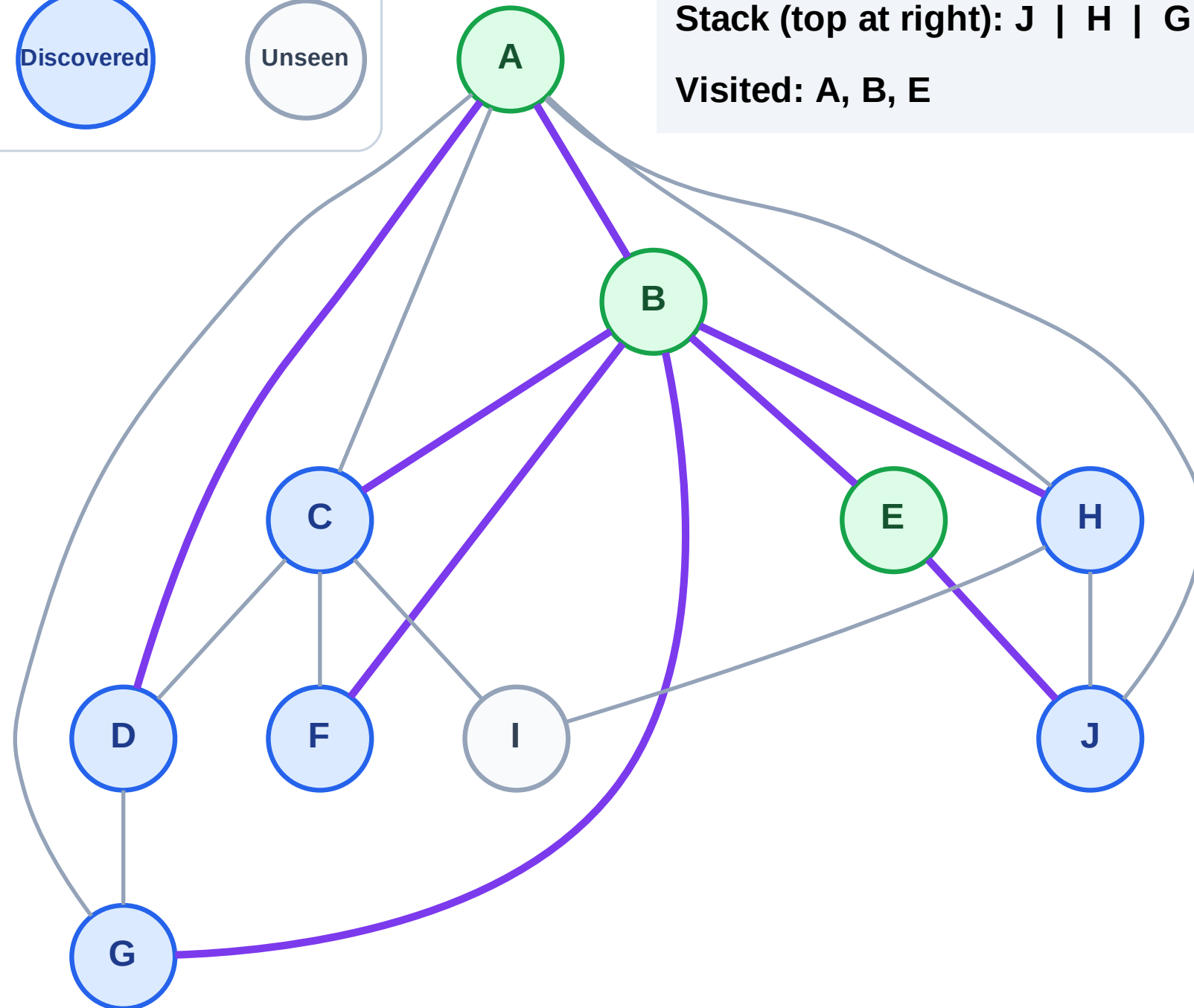
Step 7: Discover D from A

## Legend



Stack (top at right): J | H | G | F | C | D

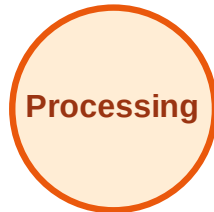
Visited: A, B, E



# DFS Traversal

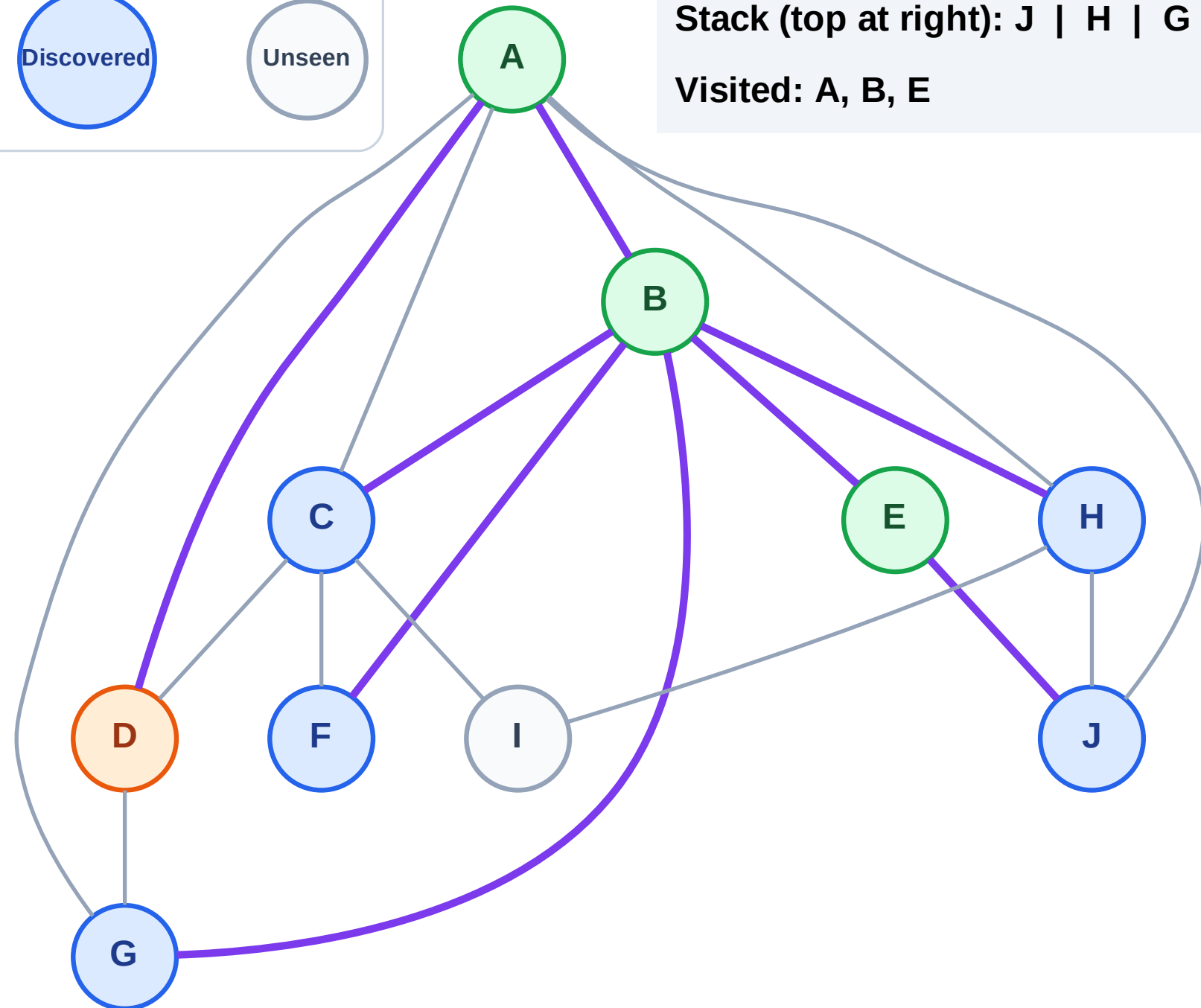
Step 8: Process node D

## Legend



Stack (top at right): J | H | G | F | C

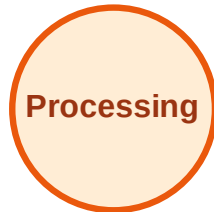
Visited: A, B, E



# DFS Traversal

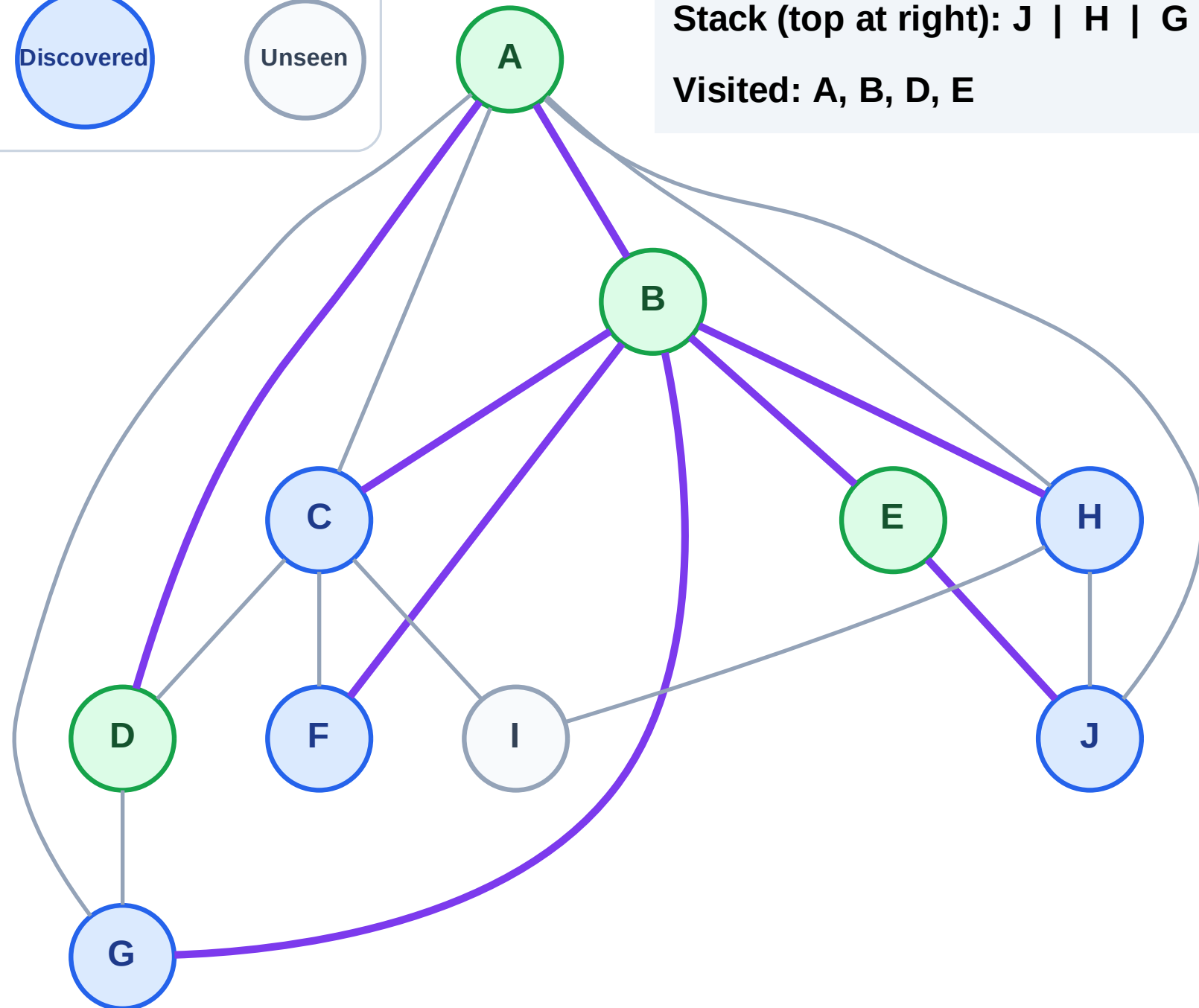
Step 9: No new neighbors from D

## Legend



Stack (top at right): J | H | G | F | C

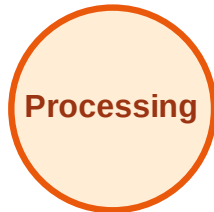
Visited: A, B, D, E



# DFS Traversal

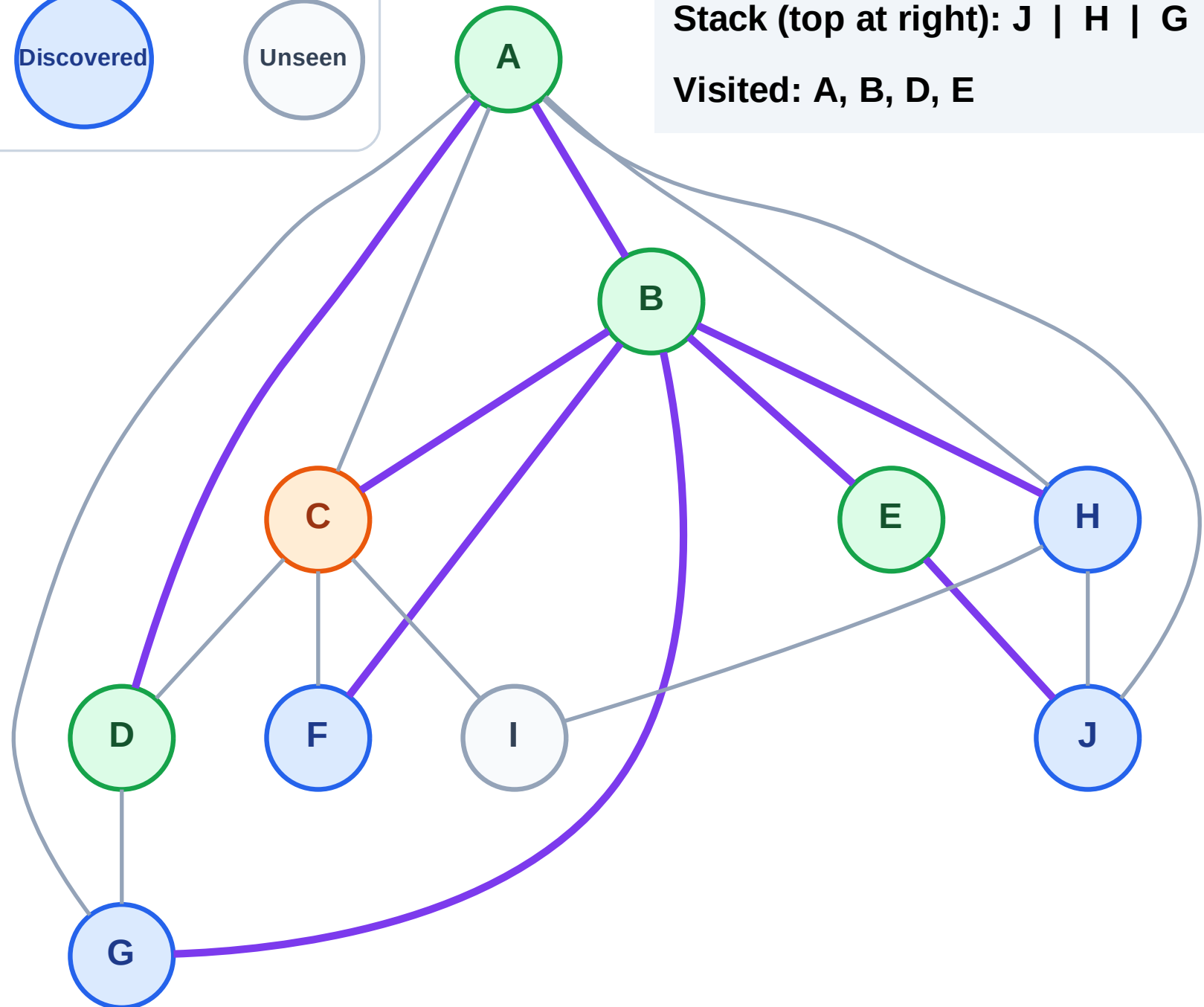
Step 10: Process node C

## Legend



Stack (top at right): J | H | G | F

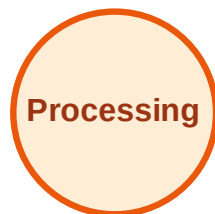
Visited: A, B, D, E



# DFS Traversal

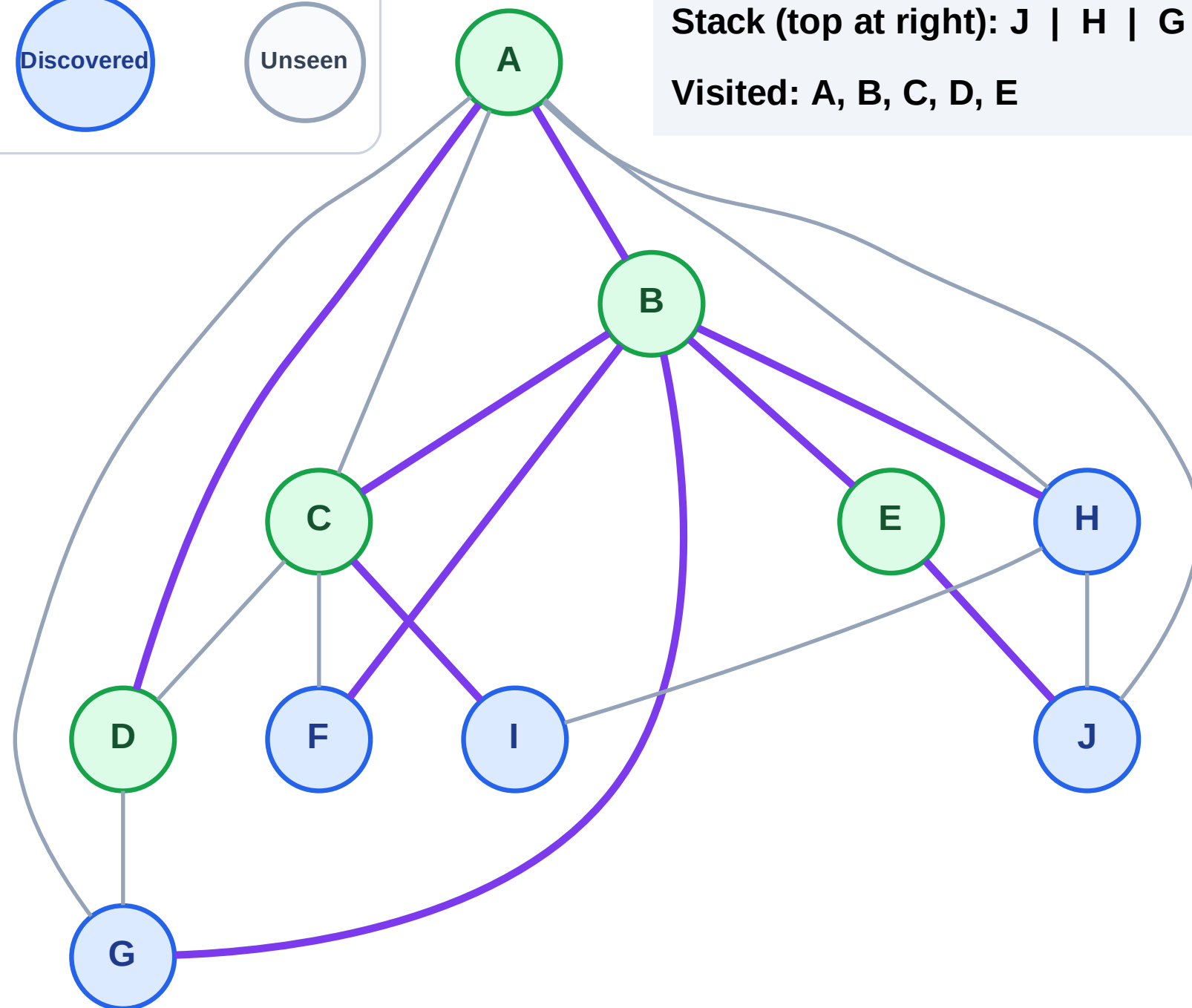
Step 11: Discover I from C

## Legend



Stack (top at right): J | H | G | F | I

Visited: A, B, C, D, E





# DFS Traversal

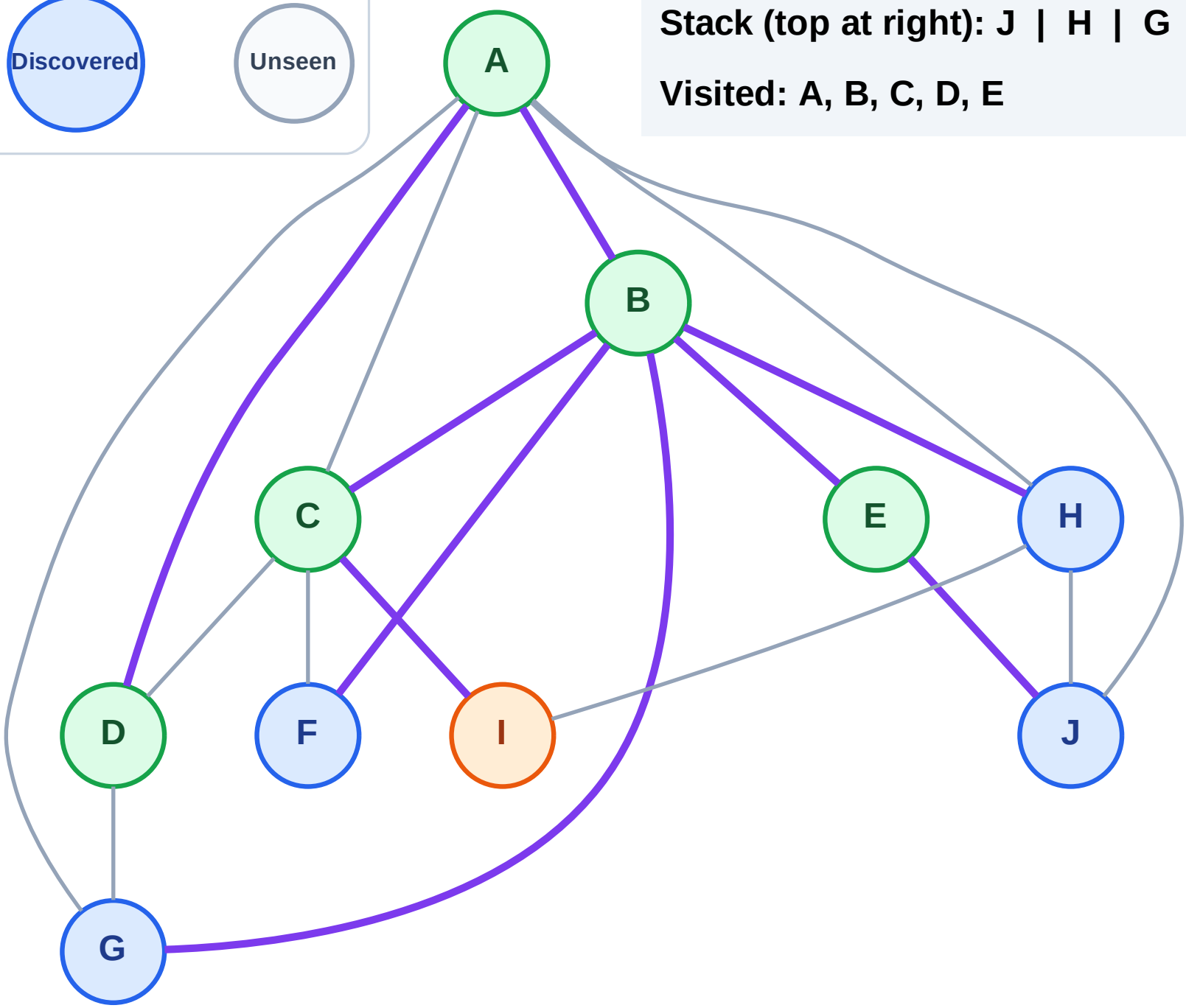
Step 12: Process node I

## Legend



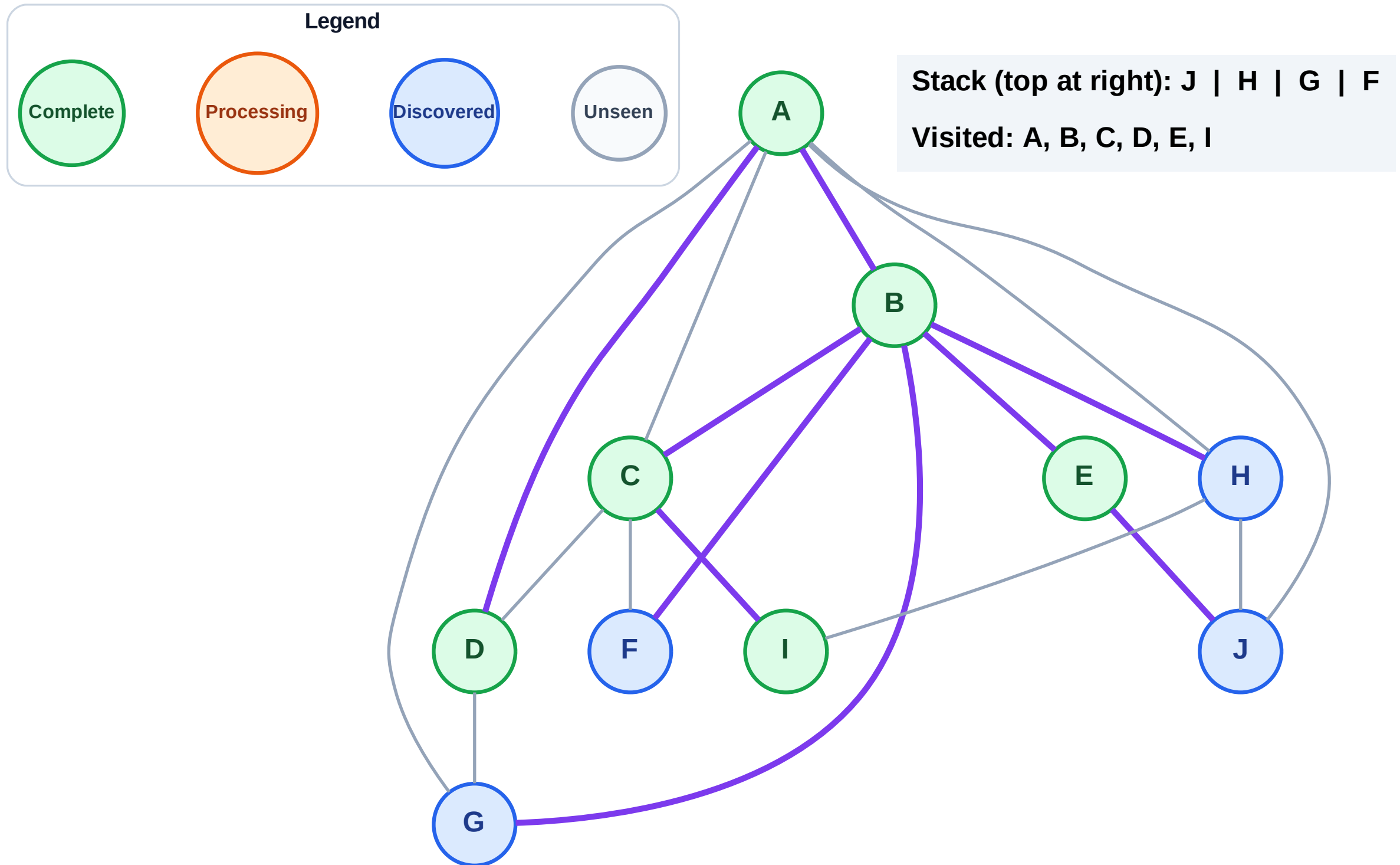
Stack (top at right): J | H | G | F

Visited: A, B, C, D, E



Step 13: No new neighbors from I

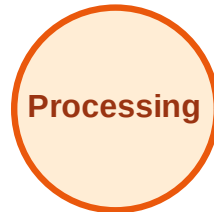
Step 13: No new neighbors from I



# DFS Traversal

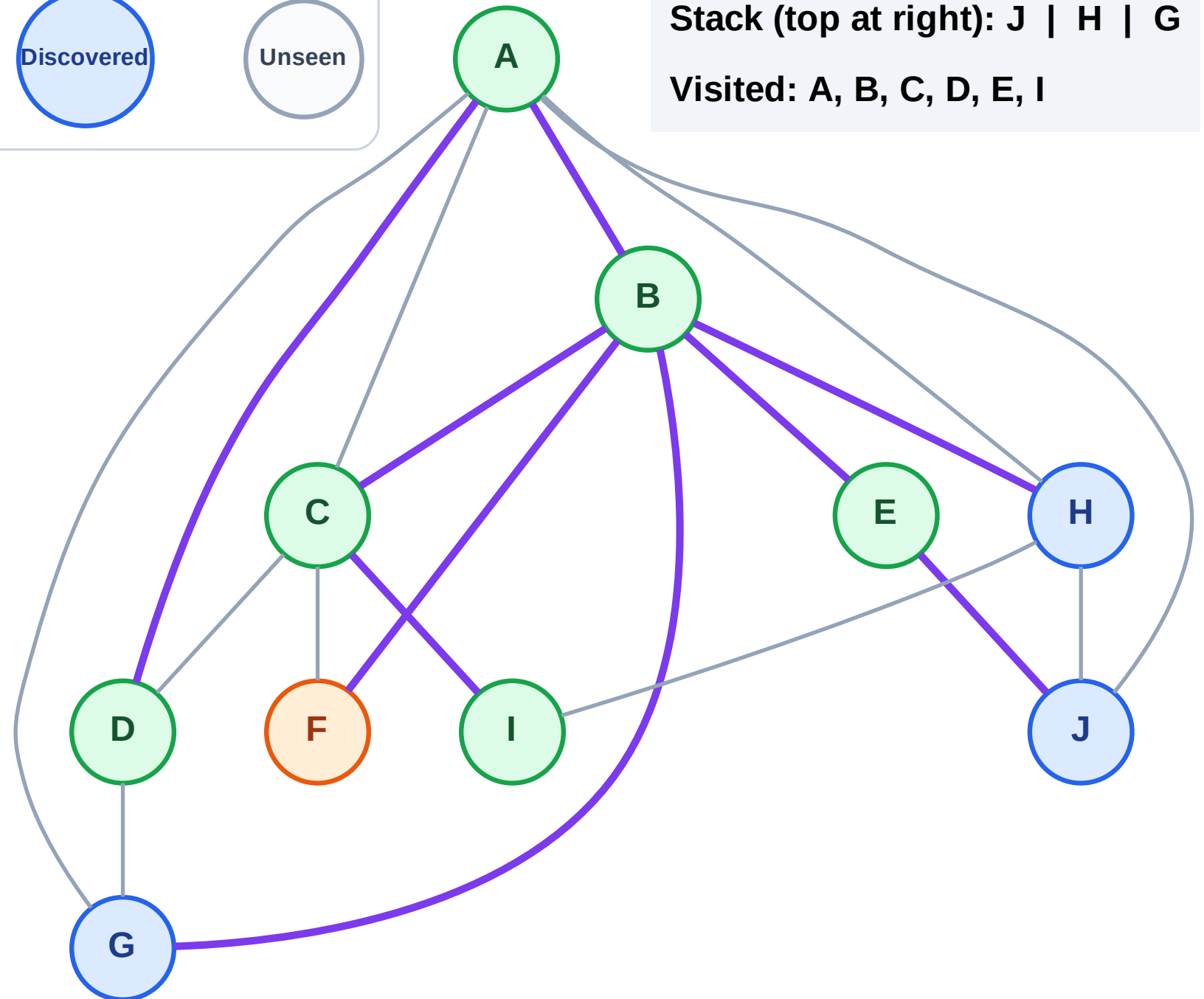
Step 14: Process node F

## Legend



Stack (top at right): J | H | G

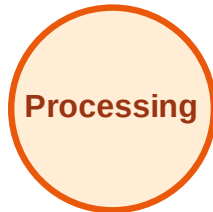
Visited: A, B, C, D, E, I



# DFS Traversal

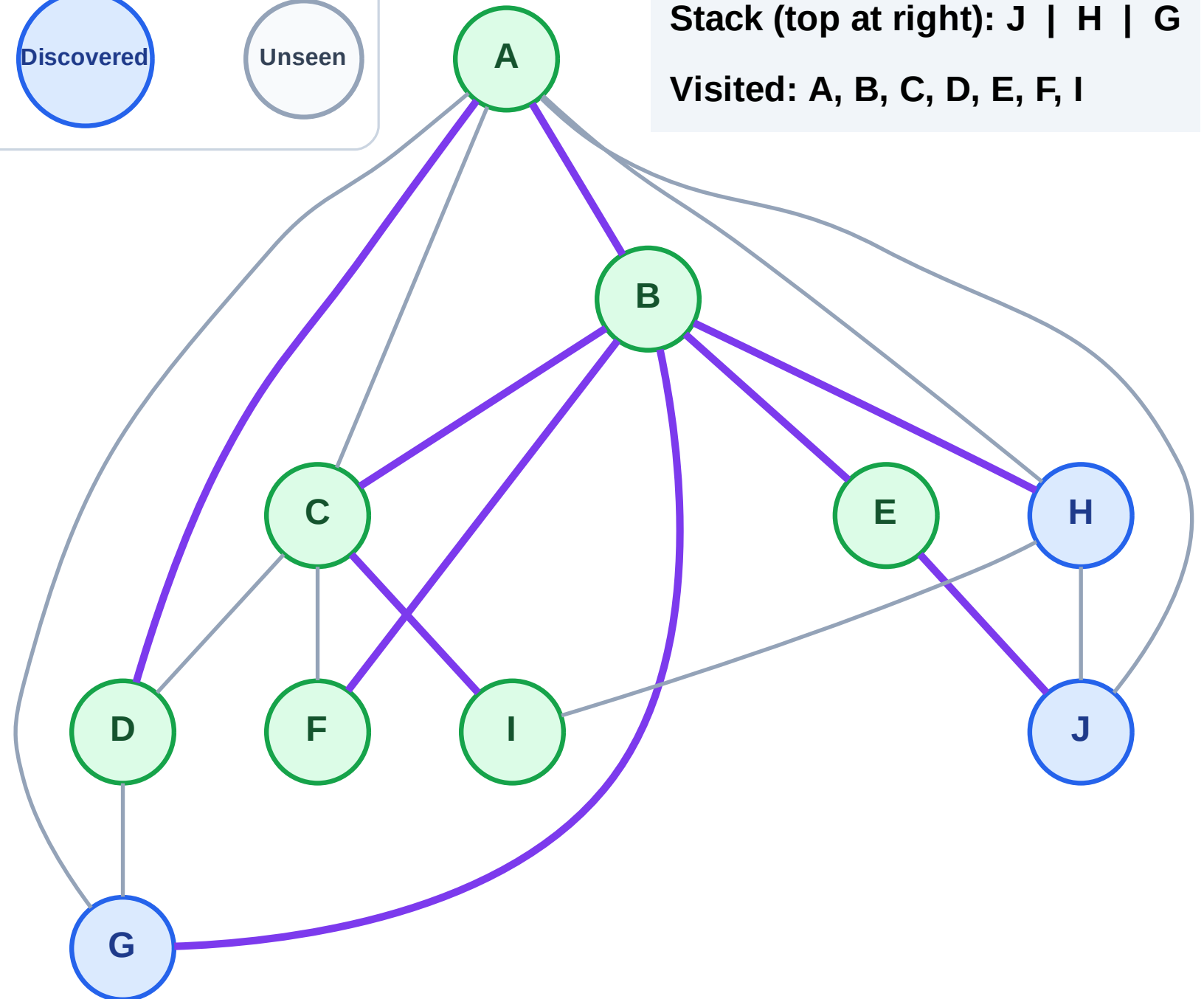
Step 15: No new neighbors from F

## Legend



Stack (top at right): J | H | G

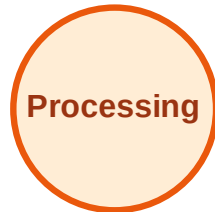
Visited: A, B, C, D, E, F, I



# DFS Traversal

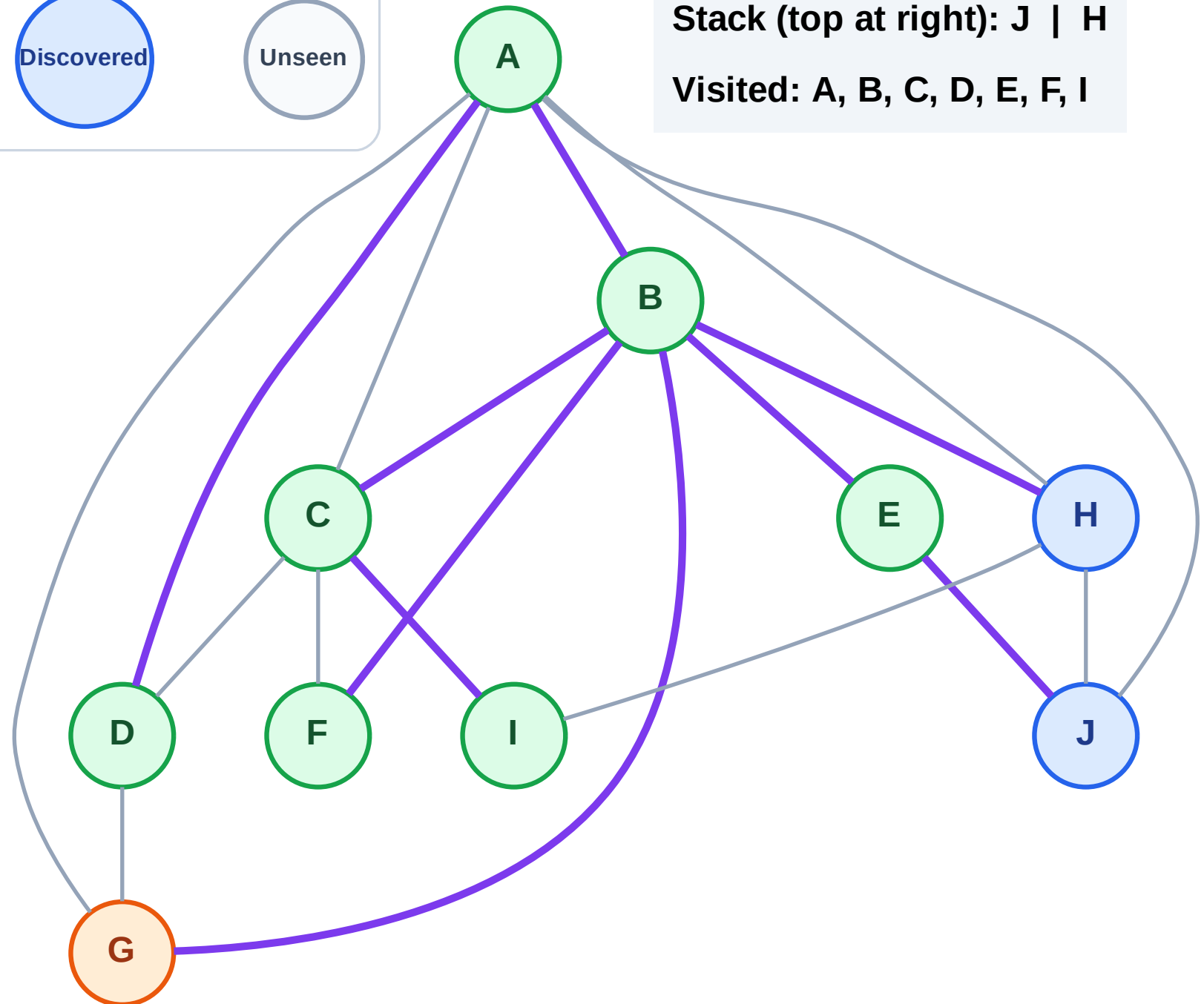
Step 16: Process node G

## Legend



Stack (top at right): J | H

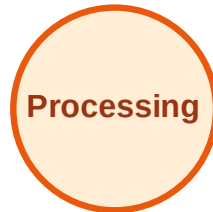
Visited: A, B, C, D, E, F, I



# DFS Traversal

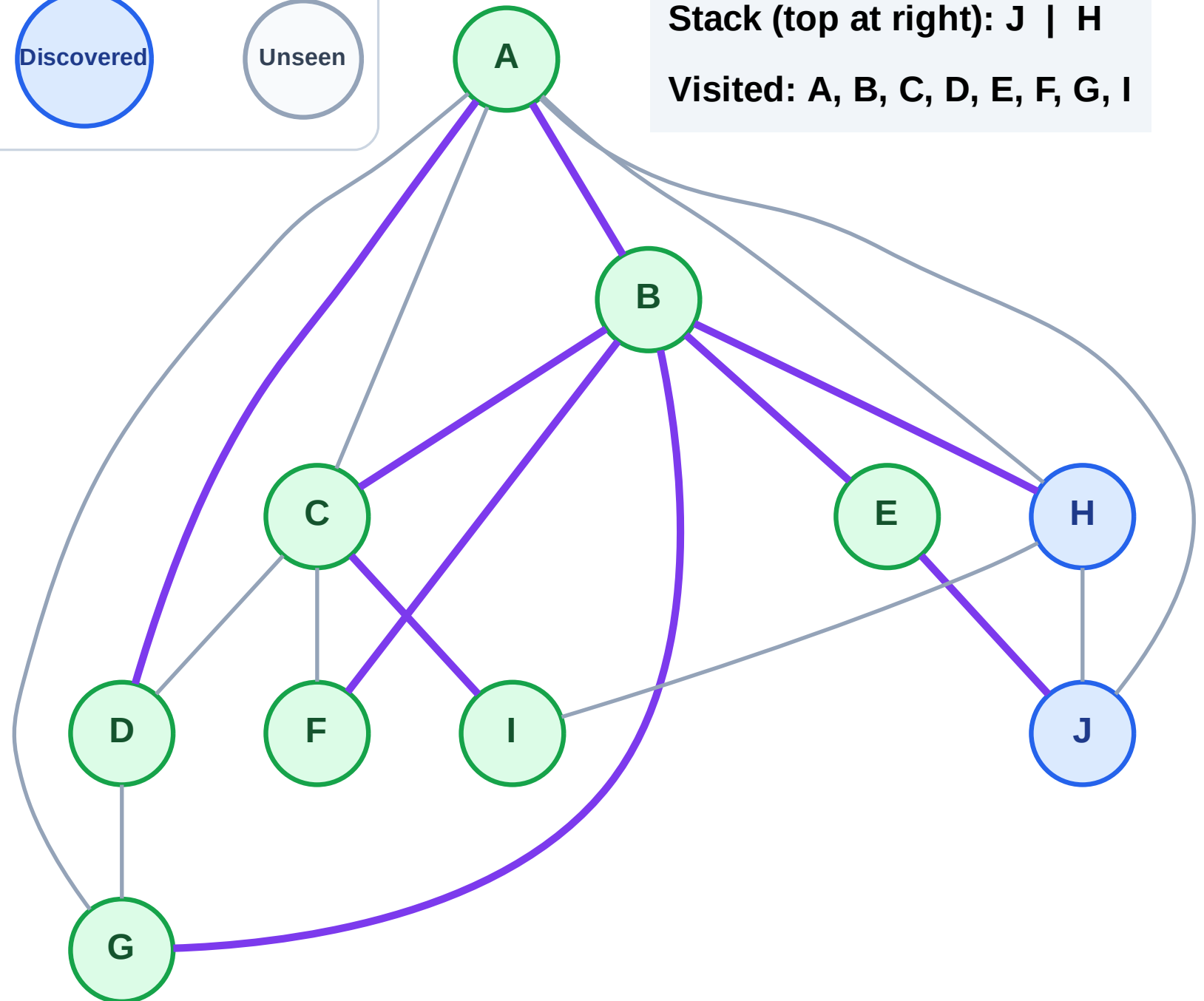
Step 17: No new neighbors from G

## Legend



Stack (top at right): J | H

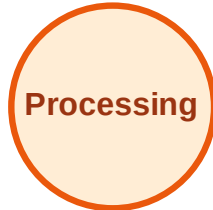
Visited: A, B, C, D, E, F, G, I



# DFS Traversal

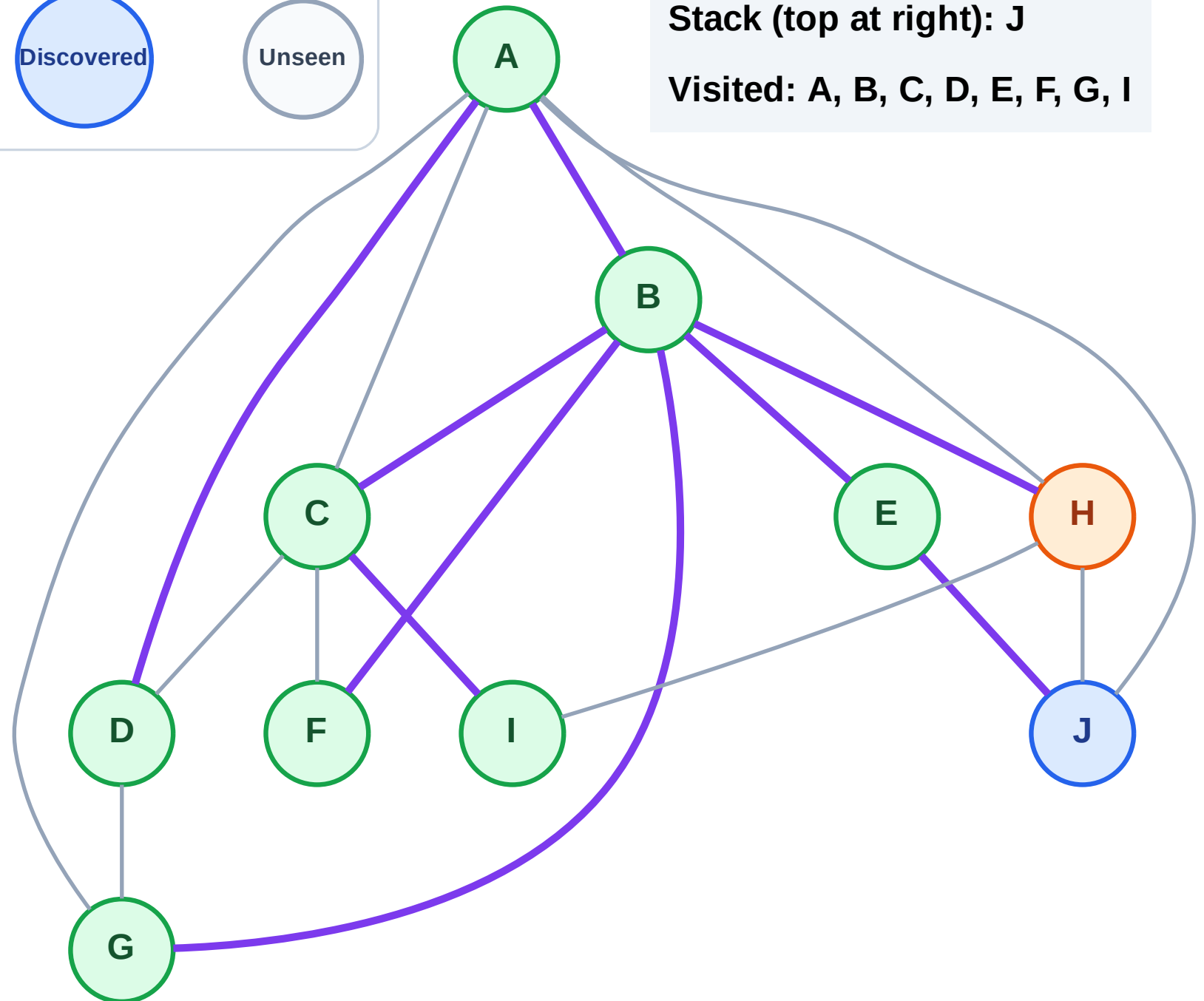
Step 18: Process node H

## Legend



Stack (top at right): J

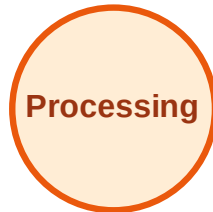
Visited: A, B, C, D, E, F, G, I



# DFS Traversal

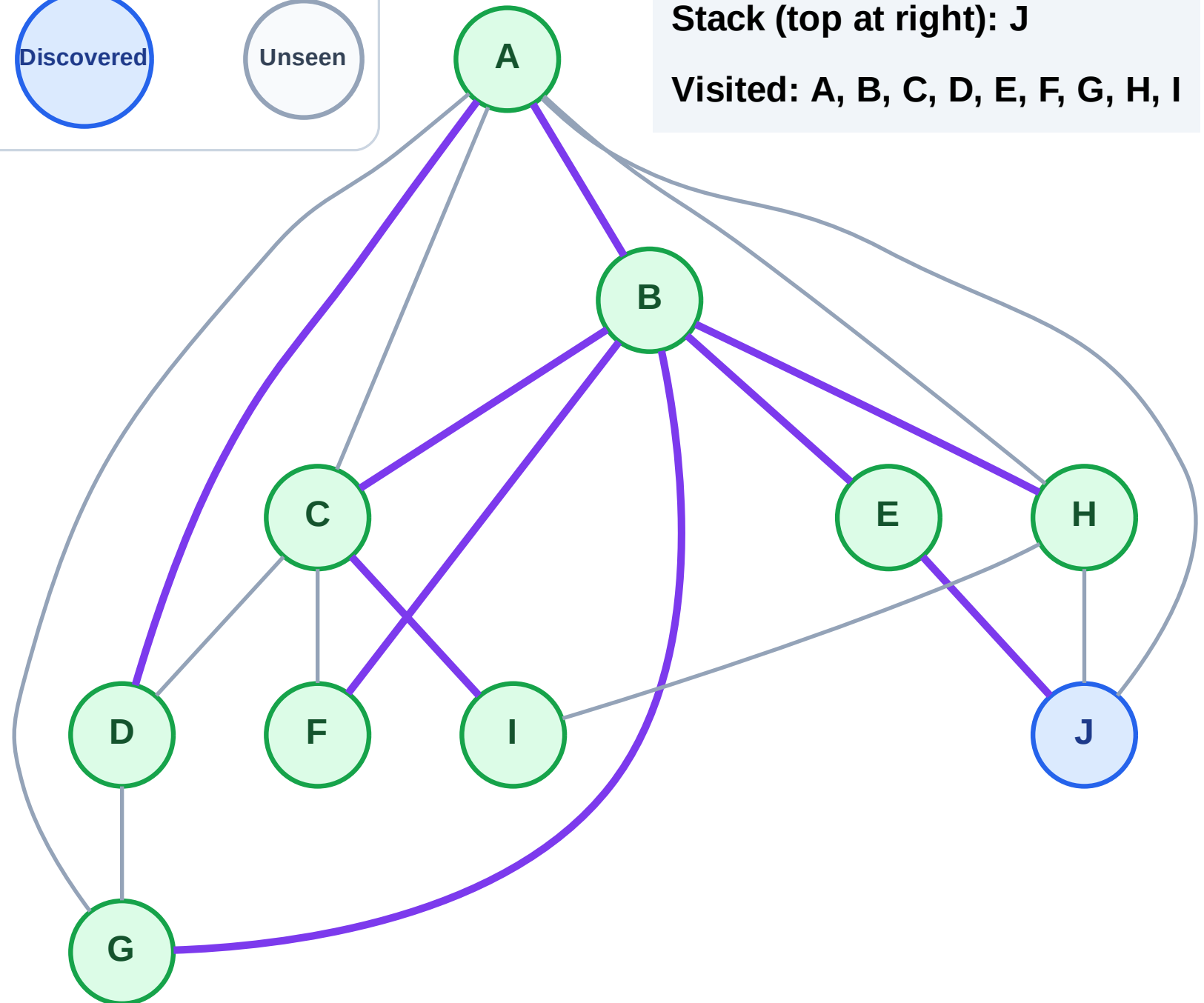
Step 19: No new neighbors from H

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I

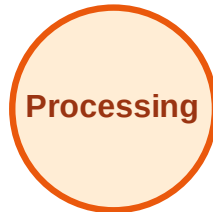




# DFS Traversal

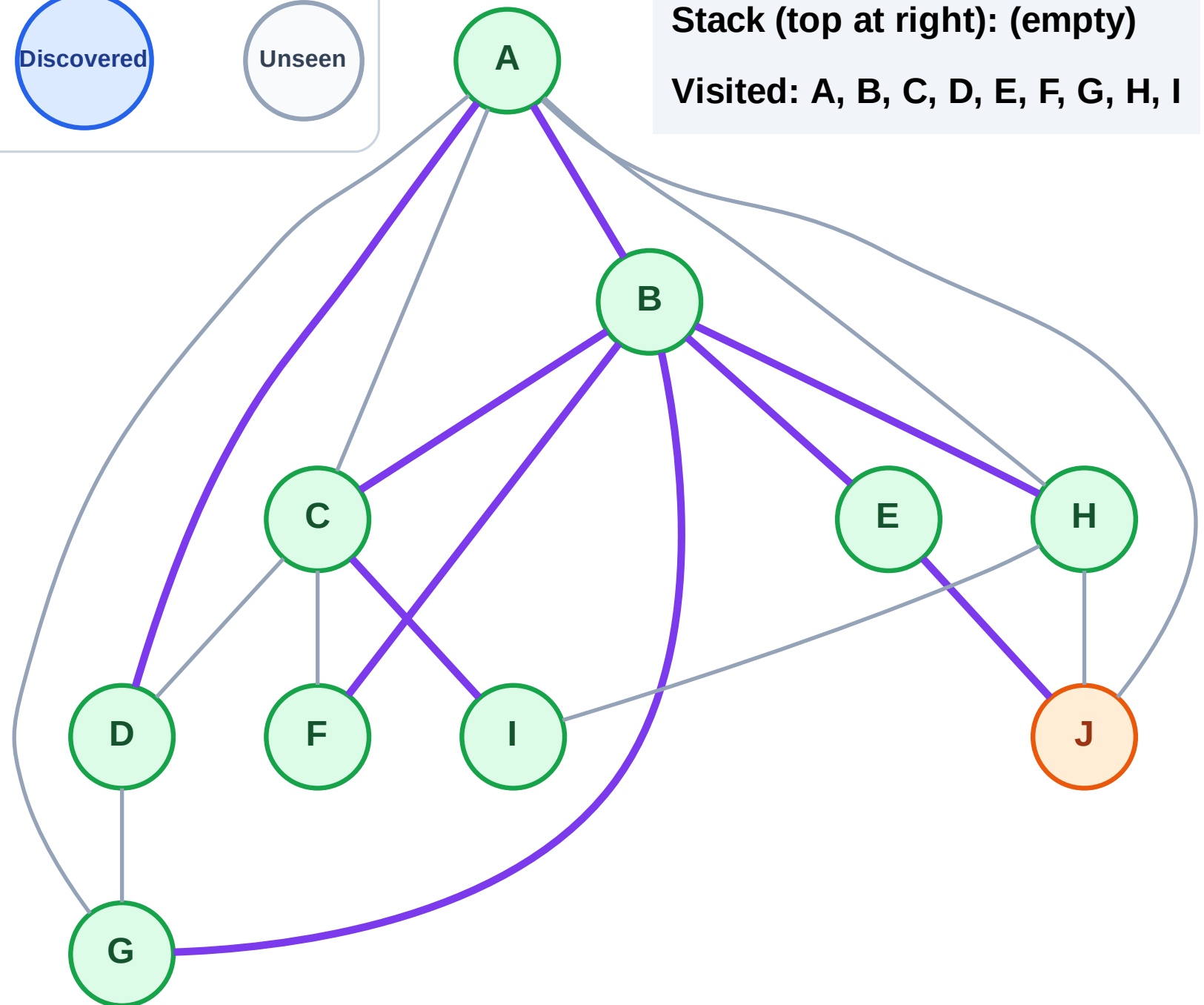
Step 20: Process node J

## Legend



Stack (top at right): (empty)

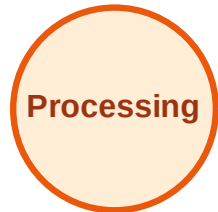
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

